

# Quality of Service

Advanced Mobile Communication Networks



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# Content

- QoS Basics
  - Requirements
  - Functions (Scheduling, Metering, etc.)
  - QoS Schemes (IntServ, DiffServ, NSIS)
- QoS over wireless
  - PHY
  - QoS-support in MAC (802.11e)
  - QoS and mobility

# QoS Basics

Questions to be answered:

- What are the QoS attributes of a connection?
- What are the basic functions for the provisioning of QoS?
- Name the strategies for QoS provisioning
- Which applications have the toughest QoS requirements?
- Name measures to minimize the error rate over wireless links
- Name the measures to minimize the average delay over wireless links
- Name the measures to maximize the throughput over wireless links
- How can you compute the maximal throughput of an end-to-end connection from its constituents?
- How compute the end-to-end delay for a connection for the case of zero transmission errors?
- What are the reasons of delay jitter, what the reasons for packet loss for wired and wireless communication?

# QoS Basics

## QoS attributes/requirements:

- data rate (throughput)
- error rate (packet loss)
- delay (latency)
- delay variation (jitter)

## Goal of QoS-enabled networks:

**Enable predictable service delivery to certain classes or types of traffic independent of other factors, e.g. other traffic or link conditions**

## Strategies (mechanisms) to ensure QoS

- **reservation** of „dedicated“ resources for a connection (e.g. CS voice, IntServ/RSVP)
- **differentiation** (e.g. prioritization) of the use of a shared resource by different connections (e.g. DiffServ)
- **overprovisioning**, i.e. dimensioning of the network such that all offered (or accepted) traffic can be handled

## Basic functions to provide QoS

- admission control (possibly including resource reservation)
- traffic classification
- traffic conditioning (traffic shaping and policing)
- scheduling
- overload control

# QoS Requirements – User (end-to-end) Requirements

Summary of applications in terms of requirements

| Acceptable error rate | Error tolerant   | Conversational voice and video                | Voice messaging                     | Streaming audio and video       | Fax                              |
|-----------------------|------------------|-----------------------------------------------|-------------------------------------|---------------------------------|----------------------------------|
|                       | Error intolerant | Telnet, interactive games distributed control | E-commerce, WWW browsing,           | FTP, still image, paging        | E-mail arrival notification      |
|                       |                  | Conversational<br>(delay $\ll 1$ sec)         | Interactive<br>(delay approx 1 sec) | Streaming<br>(delay $< 10$ sec) | Background<br>(delay $> 10$ sec) |
| Delays requirements   |                  |                                               |                                     |                                 |                                  |

# Outlook: How to Provide QoS over A Wireless Link?

## **Delay- and error-intolerant traffic** (minimize packet error rate)

⇒ Transmit it right the first time at (almost) all cost!

Employ very high SINR (high TxPower/low path loss, low interference), highly robust modulation and high FEC redundancy to ensure (almost) error free transmission

## **Delay-intolerant, error-tolerant traffic** (aim for small bit error rate)

⇒ Try to do it right the first time!

Ensure higher SINR, employ robust modulation and some FEC redundancy to minimize frame error

## **Maximize throughput, but with higher acceptable delay**

⇒ Balance modulation, coding and SINR for max. throughput

Select (higher-order) modulation scheme and (lower) FEC redundancy that maximizes throughput (goodput) for given radio condition (rather than minimal delay or frame error rates)

Considering tradeoff between TxPower, modulation scheme and redundant coding: higher order modulation scheme increases throughput and reduces delay for burst transmissions but increases risk of errors

*We will come back to this!*

# End User Performance Requirements

## Conversational/real-time services

Source: UMTS standards

| Medium | Application                 | Degree of symmetry   | Data rate   | Key performance parameters and target values                      |                                        |                     |
|--------|-----------------------------|----------------------|-------------|-------------------------------------------------------------------|----------------------------------------|---------------------|
|        |                             |                      |             | End-to-end<br>One-way<br>Delay                                    | Delay<br>Variation<br>within<br>a call | Information<br>loss |
| Audio  | Conversational voice        | Two-way              | 4-25 kb/s   | <150 msec preferred<br><400 msec limit                            | < 1 msec                               | < 3% FER            |
| Video  | Videophone                  | Two-way              | 32-384 kb/s | < 150 msec preferred<br><400 msec limit<br>Lip-synch : < 100 msec |                                        | < 1% FER            |
| Data   | Telemetry - two-way control | Two-way              | <28.8 kb/s  | < 250 msec                                                        | N.A                                    | Zero                |
| Data   | Interactive games           | Two-way              | < 1 KB      | < 250 msec                                                        | N.A                                    | Zero                |
| Data   | Telnet                      | Two-way (asymmetric) | < 1 KB      | < 250 msec                                                        | N.A                                    | Zero                |

**FER: Frame Error Rate**

# End User Performance Requirements

## Interactive services

| Medium | Application                                               | Degree of symmetry | Data rate | Key performance parameters and target values |                 |                  |
|--------|-----------------------------------------------------------|--------------------|-----------|----------------------------------------------|-----------------|------------------|
|        |                                                           |                    |           | One-way Delay                                | Delay Variation | Information loss |
| Audio  | Voice messaging                                           | Primarily one-way  | 4-13 kb/s | < 1 sec for playback<br>< 2 sec for record   | < 1 msec        | < 3% FER         |
| Data   | Web browsing - HTML                                       | Primarily one-way  |           | < 4 sec /page                                | N.A             | Zero             |
| Data   | Transaction services – high priority e.g. e-commerce, ATM | Two-way            |           | < 4 sec                                      | N.A             | Zero             |
| Data   | E-mail (server access)                                    | Primarily One-way  |           | < 4 sec                                      | N.A             | Zero             |

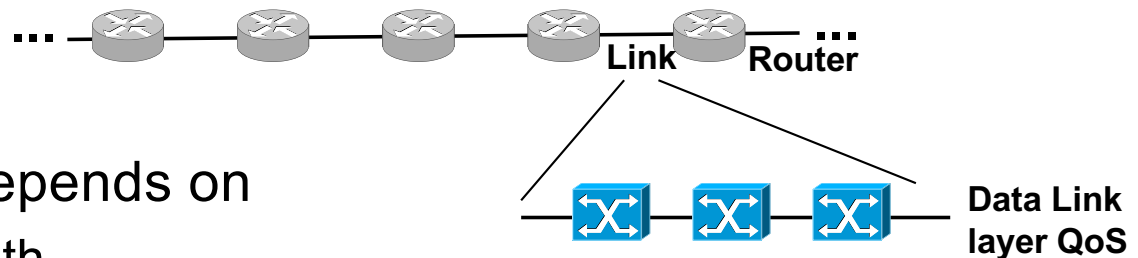
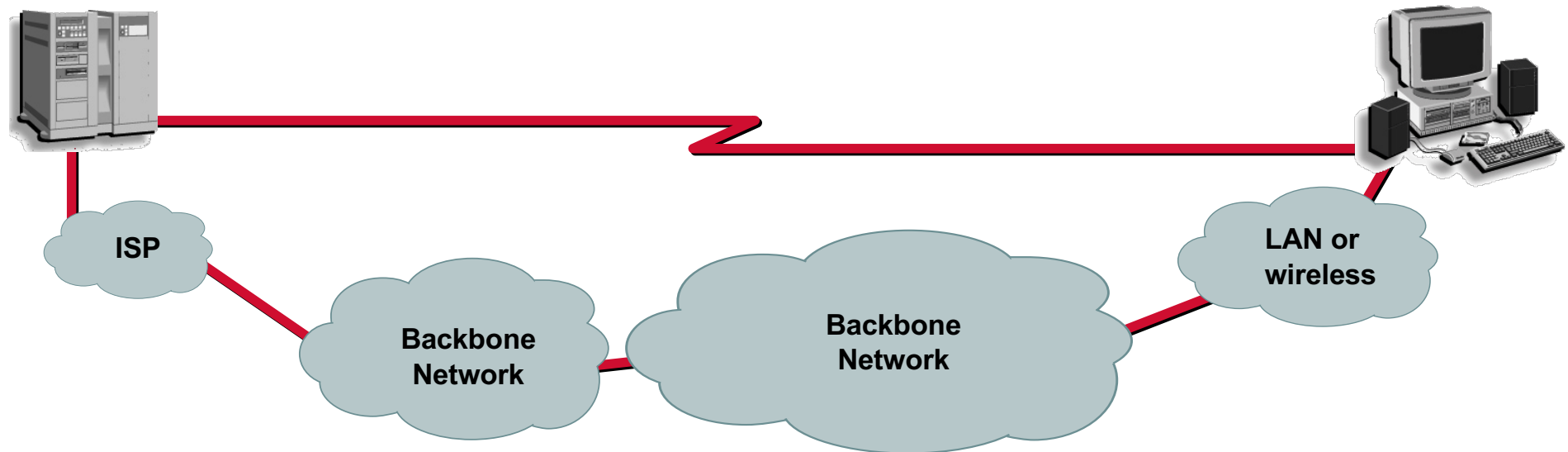


# End User Performance Requirements

## Streaming services

| Medium | Application                  | Degree of symmetry | Data rate   | Key performance parameters and target values |                 |                  |
|--------|------------------------------|--------------------|-------------|----------------------------------------------|-----------------|------------------|
|        |                              |                    |             | One-way Delay                                | Delay Variation | Information loss |
| Audio  | High quality streaming audio | Primarily one-way  | 32-128 kb/s | < 10 sec                                     | < 1 msec        | < 1% FER         |
| Video  | One-way                      | One-way            | 32-384 kb/s | < 10 sec                                     |                 | < 1% FER         |
| Data   | Bulk data transfer/retrieval | Primarily one-way  |             | < 10 sec                                     | N.A             | Zero             |
| Data   | Still image                  | One-way            |             | < 10 sec                                     | N.A             | Zero             |
| Data   | Telemetry - monitoring       | One-way            | <28.8 kb/s  | < 10 sec                                     | N.A             | Zero             |

# QoS in Networks – End-to-end QoS



- Network-layer QoS depends on
  - routers along the path
  - characteristics of each link's technology (layer 1 and 2)

# Edge-to-edge QoS in Wired Systems

- Latency

Processing delays experienced within each router

Transmission delays across each link (fairly predictable)

- Jitter

Introduced within routers by unrelated traffic passing through **shared resources** at congestion points (queueing delays)

- Packet loss

Routers provide only **finite buffering capacity** (congestion points)



**But: in wireless, the radio link is the most crucial part for QoS**

## QoS on Wireless Link

- Latency
  - Processing esp. for channel coding
  - Neglectable transmission delay
- Jitter
  - Scheduling: wait for appropriate time slot
  - Packet loss (and time for local retransmission)
  - System load, i.e. dependency on other traffic
  - Channel quality in case of opportunistic scheduling (exploiting differences between channel quality of different mobiles)
- Packet loss
  - Unexpected deterioration of link quality (maladjusted modulation, coding, TxPower) or below minimum SINR or signal strength requirements

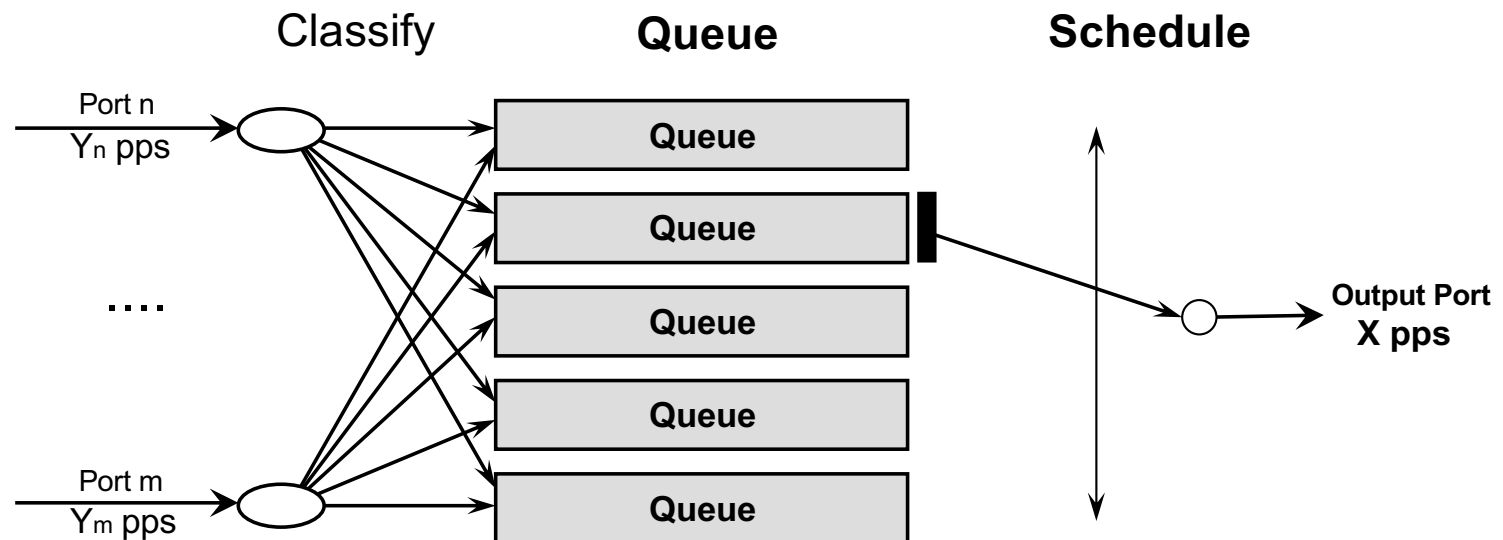
# QoS Functions and Common Concepts

Questions to be answered:

- Which functions in a QoS-enabled network ensure that an application does not exceed its throughput requirements?
- How does the token bucket scheme help in the limitation of the user throughput? How does the system support averaging?
- What is the purpose of explicit packet dropping in routers?
- How influences packet dropping the performance of TCP connections?
- What are the ingredients of the TOS field? Why are multiple fields needed?
- Why is the TOS field mostly relevant to the network as such rather than the end system?
- Why do networks differentiate between edge routers and core router? What's the purpose and what is the difference in behavior?

# Basic QoS Functions – Scheduling

- Classification of packets (Traffic classes)
- A queue for each class of traffic
  - Queue management
  - Different packet discard functions
- Queues must share finite capacity of output link → scheduler



# Basic QoS Functions – Traffic Shaping and Policing

- Traffic Shaping
  - Delay packets exceeding the specified throughput to ensure compliance with the desired traffic profile
  - Typically implemented by the application
- Policing
  - Monitoring of traffic for compliance with its traffic profile and taking respective measures if not, e.g. dropping or marking as non-compliant
- Marking

Packets are marked if they exceed a burstiness threshold

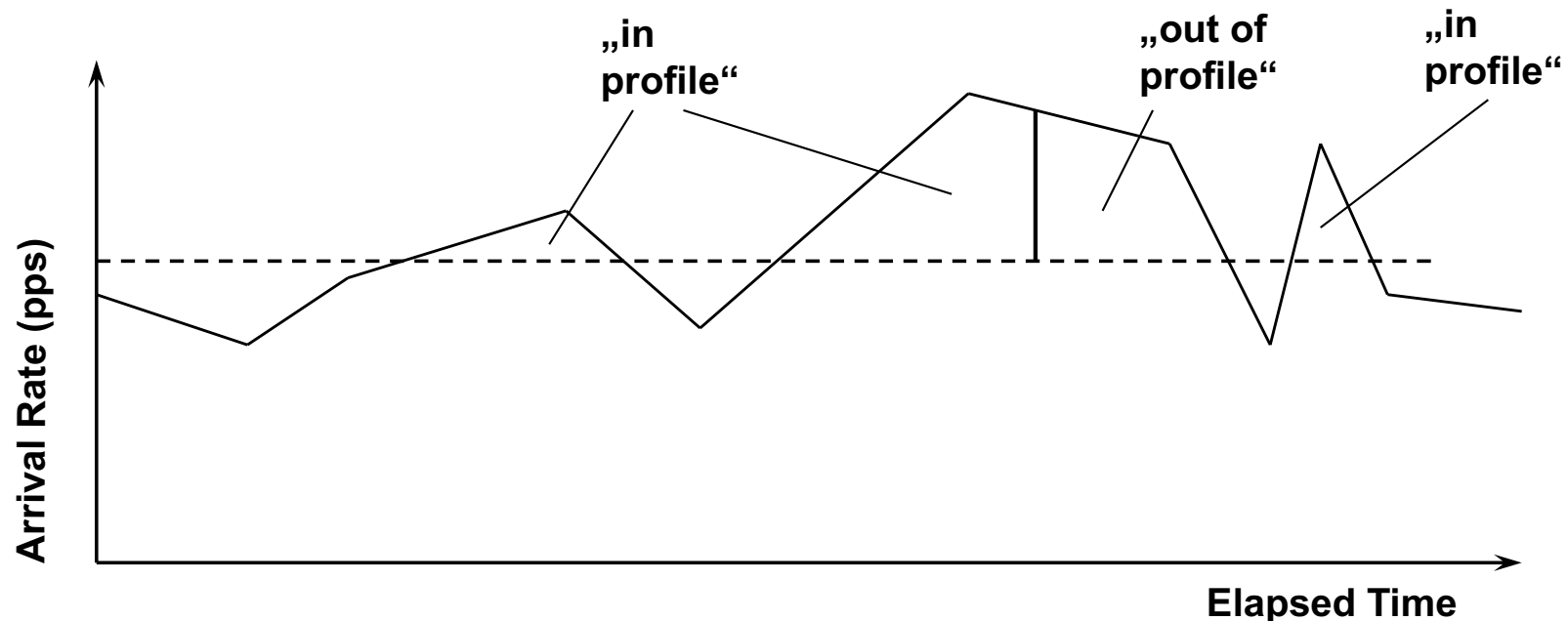
  - The system can schedule such packets with lower priority
  - In case of congestion, marked packets are dropped first
- Reordering
  - Within one queue unmarked packets are scheduled before marked ones

# Metering

Policing and Marking share a common component – a metering function detecting whether a packet is „in“ or „out of profile“

Example: Token Bucket Meter

- Allows a small degree of burstiness
- Enforces a lower average rate limit





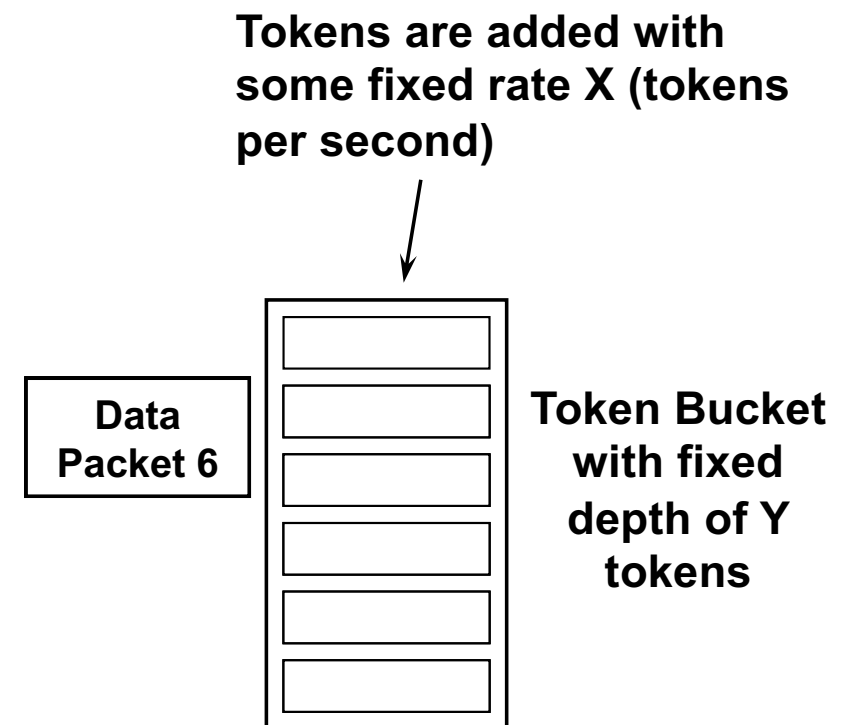
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Example: Token Bucket Meter

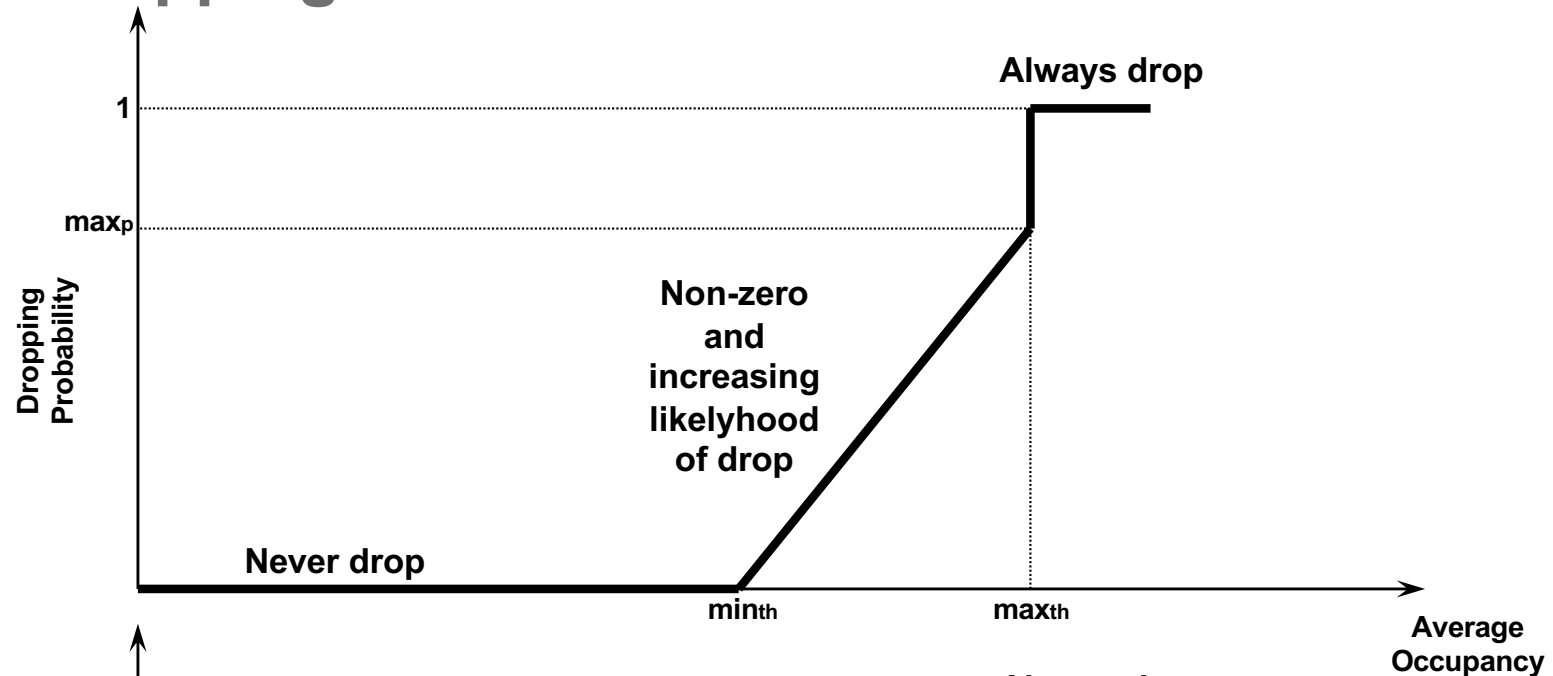
Whenever a packet arrives,  
one token is removed from the bucket  
and the packet is marked to be “in profile”

Whenever a packet arrives and  
no token is available in the bucket,  
the packet is marked to be “out of profile”

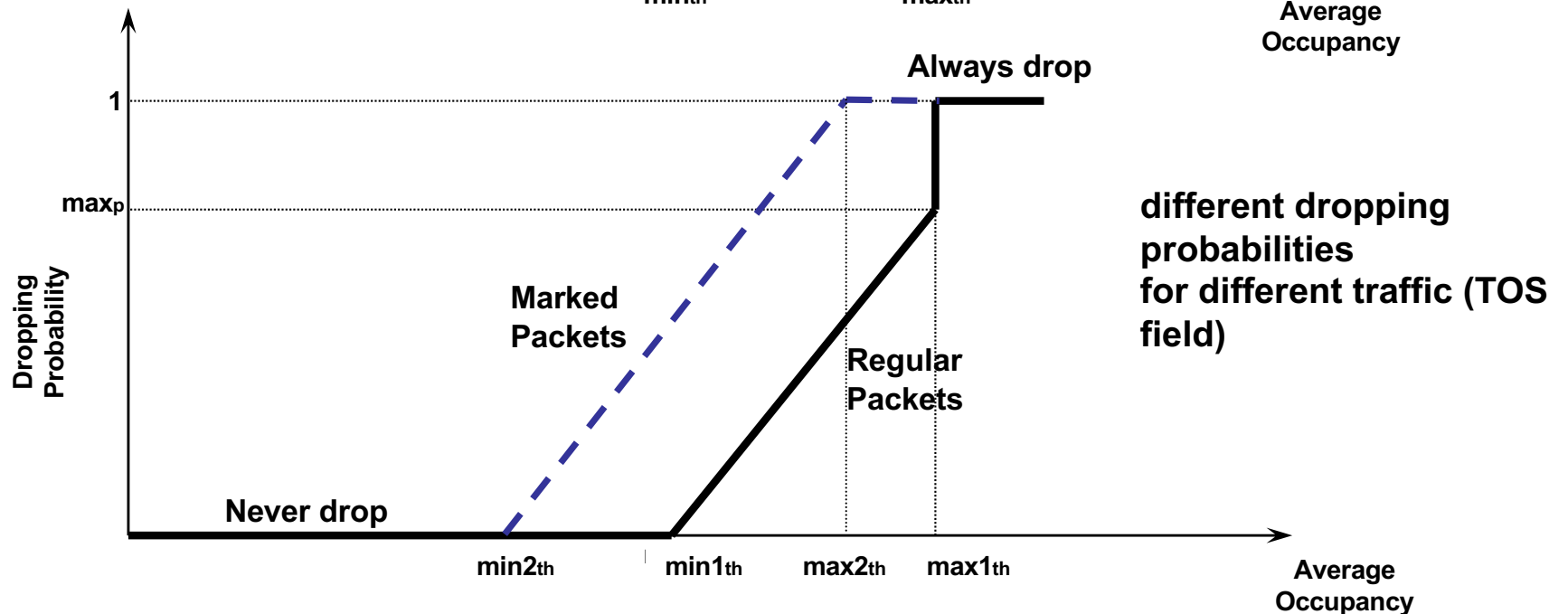


# Packet Dropping in Routers

## Random Early Detection (RED)

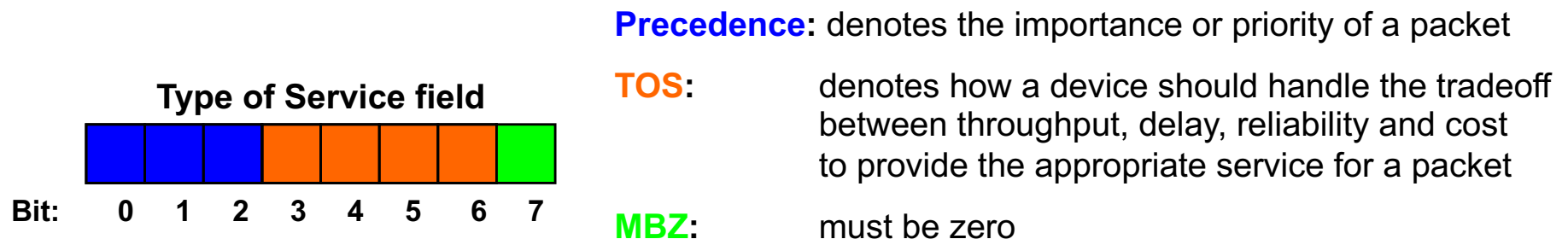


## Weighted Random Early Detection



## QoS in IP Networks – IP Packet Marking (TOS Field)

- Packet Marking assigns a priority level to each packet
- Devices supporting traffic prioritisation can use this information to provide traffic shaping capabilities enabling QoS
- In IP-based networks this priority level is stored in the Type of Service (TOS) field (8 bits) of the IP header:



- There are multiple interpretations of the TOS field in the IP header
- Here we follow RFC 1349
- See [https://en.wikipedia.org/wiki/Type\\_of\\_service](https://en.wikipedia.org/wiki/Type_of_service) for details

# QoS Strategies and Signaling Schemes

## QoS Strategies

- Integrated Services (IntServ, or IS)
- Differentiated Services (DiffServ, or DS)
- Overprovisioning

A number of concepts are common to these QoS strategies

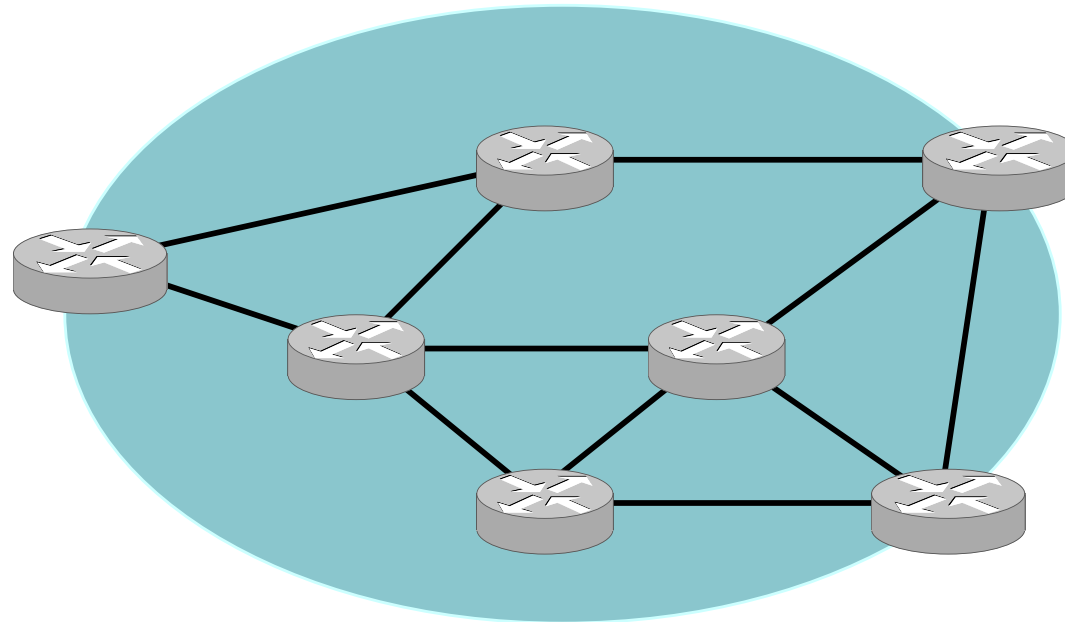
## Signaling schemes

- RSVP
- NSIS
- ...

# Common Concepts

## Network architecture comprises

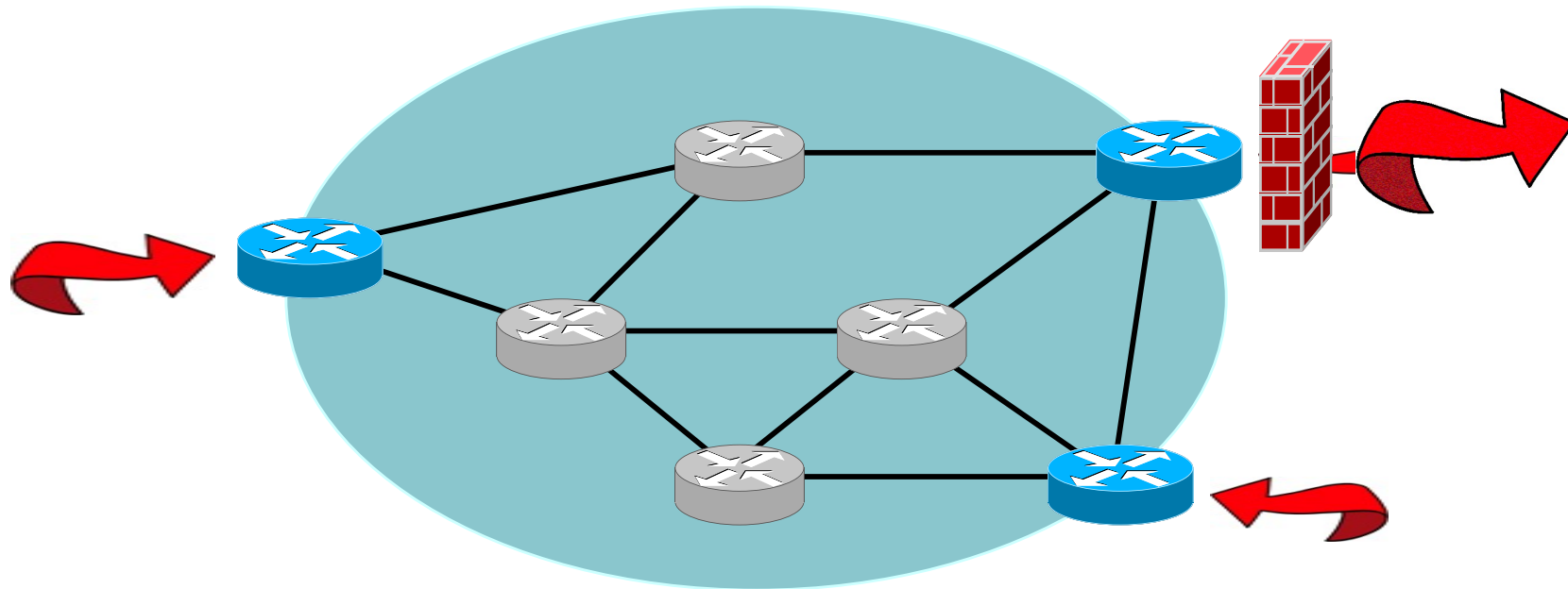
- edge routers
- core routers
- links



# Common Concepts

## Edge routers

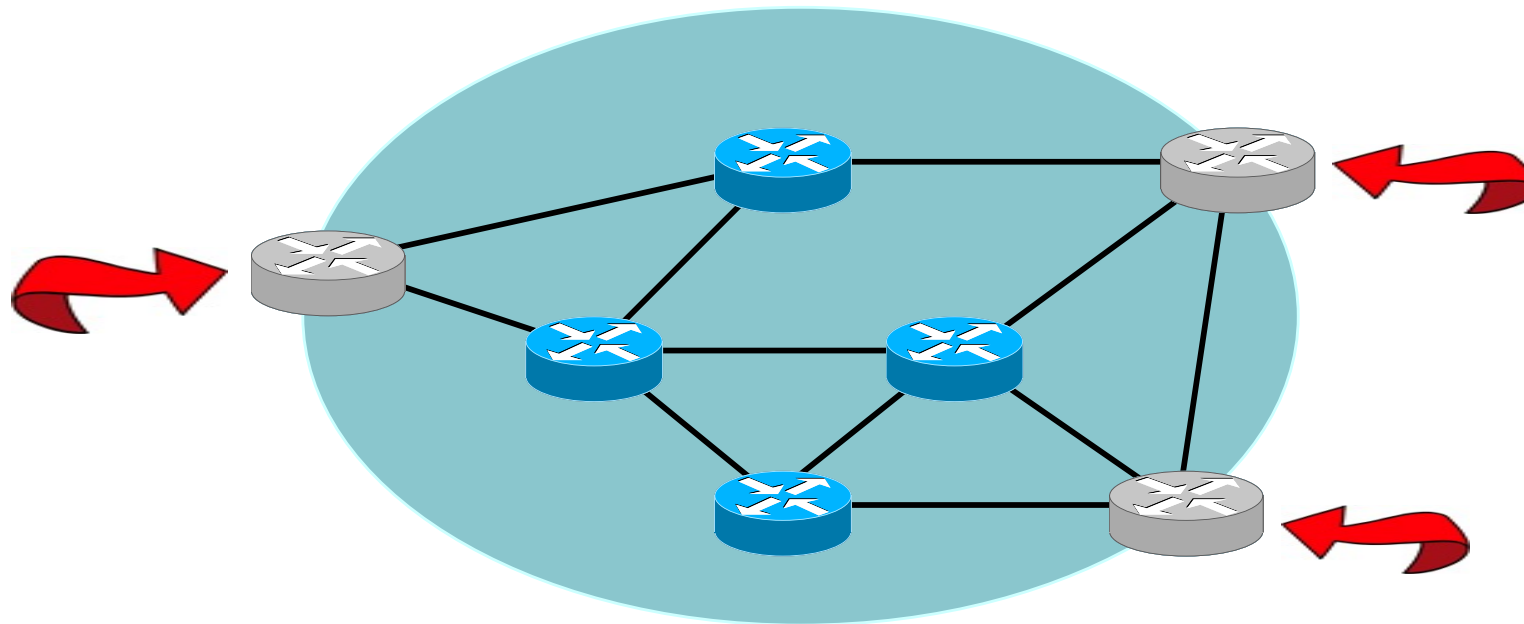
- accept customer traffic into the network
- characterize, police, and/or mark traffic, being admitted to the network
- may decline requests signaled by outside sources (admission control)



# Common Concepts

## Core routers

- provide transit packet forwarding service between other core and/or edge routers
- differentiate traffic insofar as necessary to cope with transient congestion within the network



# QoS Strategies: IntServ, DiffServ, Overprovisioning

Questions to be answered:

- What are the three main strategies for QoS provisioning in a network?
- Why does IntServ not scale? What is the problem of packet handling per individual flow? Hint: think about the implementation of rate control for individual flows.
- Why does RSVP use soft-states? What does it mean?
- How does RSVP handle the case that a router is not supporting it?
- What are the common entities in routers that are present in both IntServ and DiffServ systems?
- What are the major differences between IntServ and DiffServ wrt
  - Provided QoS requirements
  - Separation of flows
  - Effort for edge and core routers
- How can we ensure that QoS works over multiple networks and multiple providers?
- What is the difference between metering and marking?
- Under what circumstances can overprovisioning be applied?



# Integrated Services (IntServ or IS)

IntServ architecture focuses on supporting individual applications by

- per flow traffic handling at every hop along an applications end-to-end path
- an a-priori signaling of each flow's requirements (setup of the flow)

An IntServ flow (a common QoS treatment) is defined as a stream of packets with common

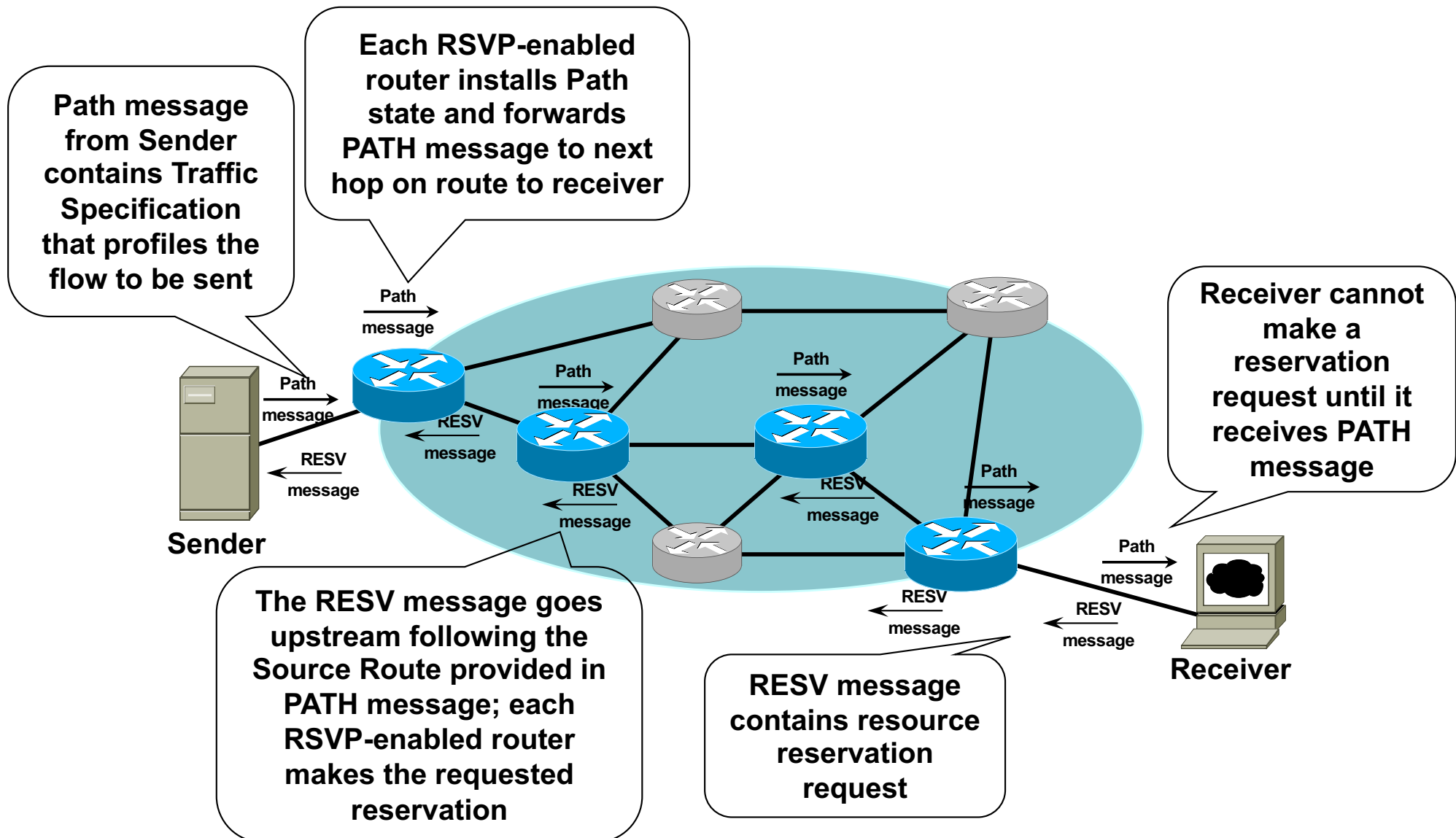
- source address,
- destination address and
- destination port (optional)

Two classes of applications are supported by IntServ:

- Real-time applications
- Traditional applications expecting a service best described as best effort „under unloaded conditions“

Signaling in the IntServ architecture to set up the flow is typically supported by the ReSerVation Protocol (RSVP)

# IntServ – Reservation Protocol



# IntServ – Reservation Protocol

RSVP is **receiver-initiated** (receiver of data flow is responsible for the initiation of the resource reservation)

RSVP is a **soft-state protocol**

- Periodic Path messages are forwarded along the routing trees provided by the routing protocol (routing from source to sinks based on regular IP mechanism)
- Reservation refresh messages are forwarded along the sink trees (based on state information maintained by each router) to maintain current reservation state (identical to initial resv request)

RSVP supports **heterogeneous receivers in a multicast group**

- multicast group membership may change dynamically  
→ reservation must be renewed (soft-state protocol)
- multicast group members „switch channels“

[Compare to sender-initiated approach: the sender would be responsible for resource reservation for all multicast group members!]

# IntServ – Summary

## Pros

- Provides the highest possible level of QoS

## Cons

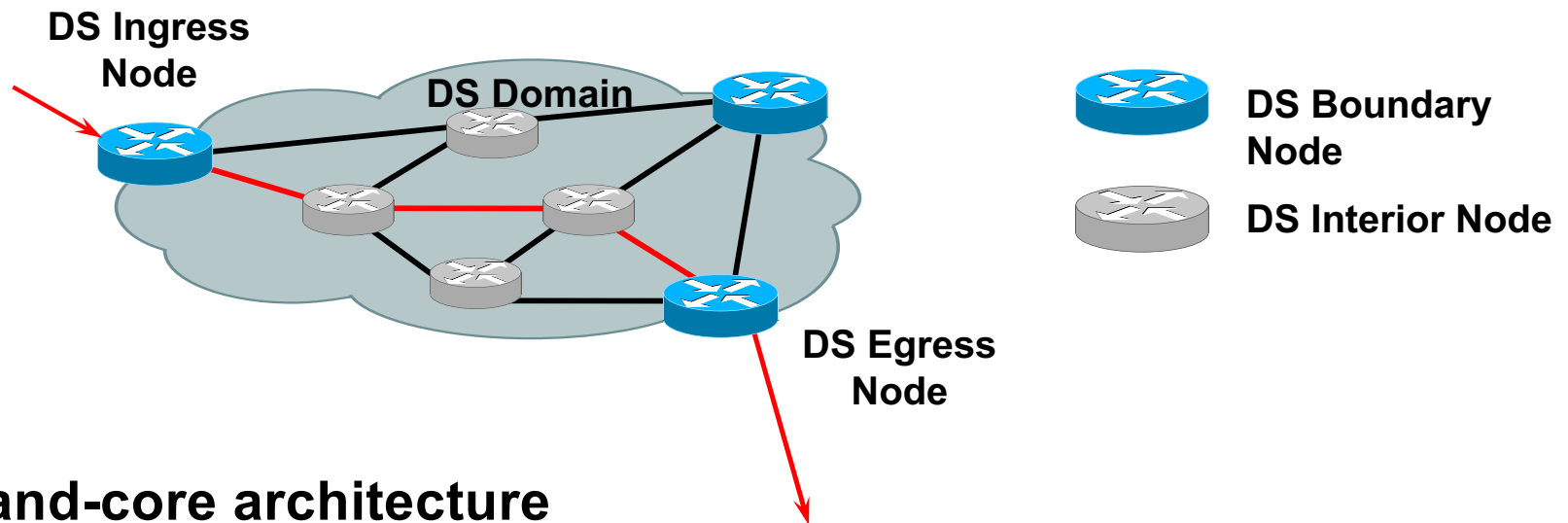
- Scalability problem: each flow, i.e. each IP packet, must be handled and maintained by each router on the data path individually (consider that millions of flows have to be managed by a core router)
- Signaling overhead due to RSVP soft-state behavior
- Shortest path routing (OSPF) may not be optimal
- No fairness, i.e. fair distribution of limited resources among aspirants
- Violation of IP principle to keep individual states of connections in the edges (hosts) only

# Differentiated Services (DiffServ, or DS)

**Idea: handle larger traffic entities in a common way rather than each flow individually**

- alternative to the high complexity of the IntServ architecture
- incremental improvements on the best-effort service model
- remove complexity from the core nodes => scalability

## Terminology

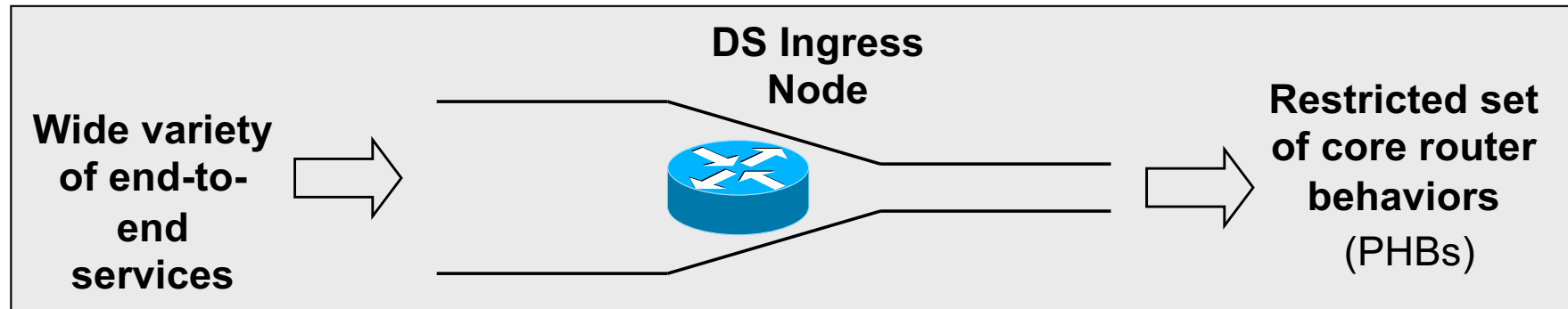


## Edge-and-core architecture

- complex decision making is pushed to the edges
- edge-to-edge services are built from a small set of core router behaviors

# DiffServ – Traffic Classification

Edge-and-core architecture requires mapping of a wide variety of traffic into a **restricted set of core router behaviors** within the DS Ingress Node



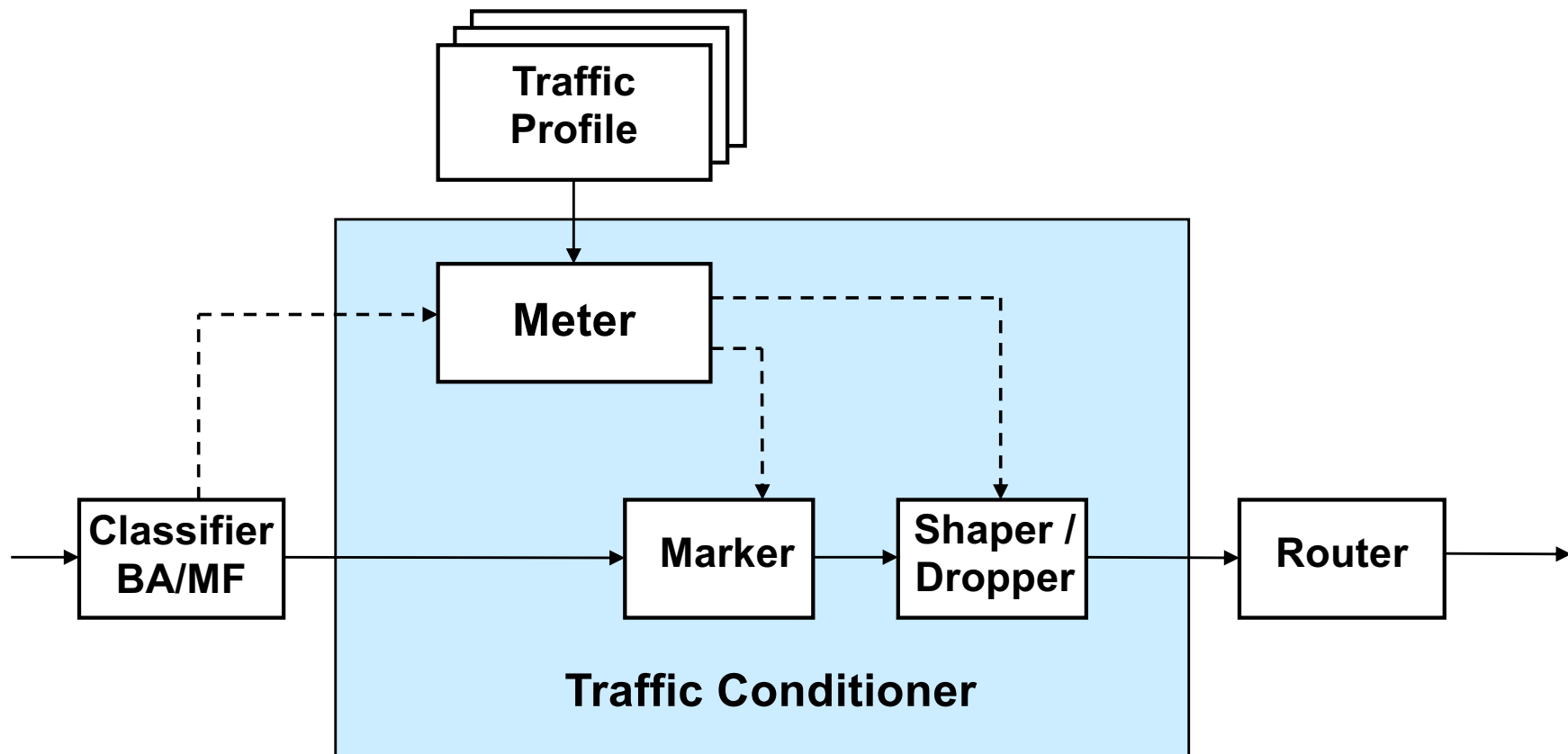
Two primary types of **DiffServ classifiers** (applied in ingress node):

- Behavior Aggregate (BA)
  - packet classification solely based on DiffServ (DS) field (Differentiated Services Code Point – DSCP values) in IP header (former TOS field)
- Multi-Field (MF)
  - packet classification based on multiple fields of the header, e.g.
    - source and destination addresses
    - source and destination ports
    - protocol ID

Within a DiffServ domain many microflows will share a single DSCP

# DiffServ – Traffic Conditioning

- Metering
  - monitoring if traffic meets the profile (based on classification)
- Marking
  - setting of the DS field
- Shaper/dropper
  - queueing
  - priority degradation or dropping where negotiated rate is exceeded



# DiffServ – Per-hop Behaviors (PHBs)

PHBs are a description of the externally observable forwarding behavior of a DS node applied to a particular Behavior Aggregate (BA):

- resources (buffer, bandwidth, ...)
  - priority relative to other PHBs
  - relative observable traffic characteristics (delay, loss, ...)
- no constraints with respect to implementation!

PHBs are indicated by specific values in the DSCP (TOS field)

PHBs are building blocks for edge-to-edge services

Note: DiffServ allows to map multiple DSCP values onto the same PHB

Two PHBs have been standardized by IETF:

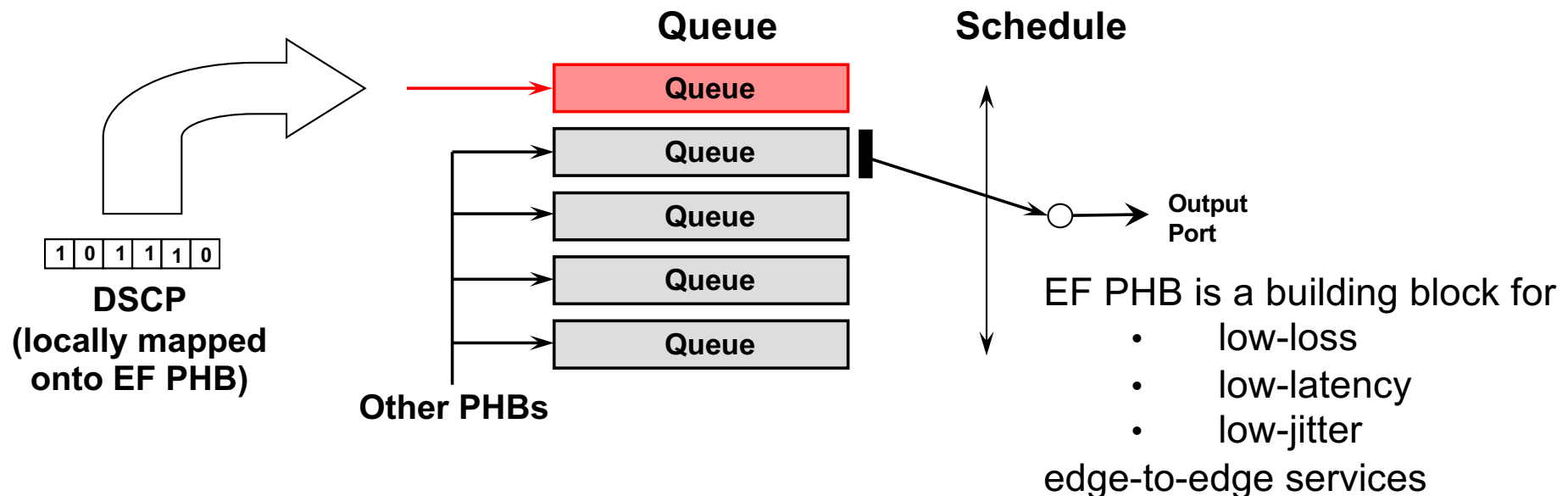
- Expedited Forwarding (EF) => premium low-loss, low-delay service!
- Assured Forwarding (AF)
- (Class Selector Per-hop Behaviors)



## DiffServ – Expedited Forwarding (EF) PHB

EF PHB requests every router along the path to service EF packets at least as fast as the rate at which EF packets arrive

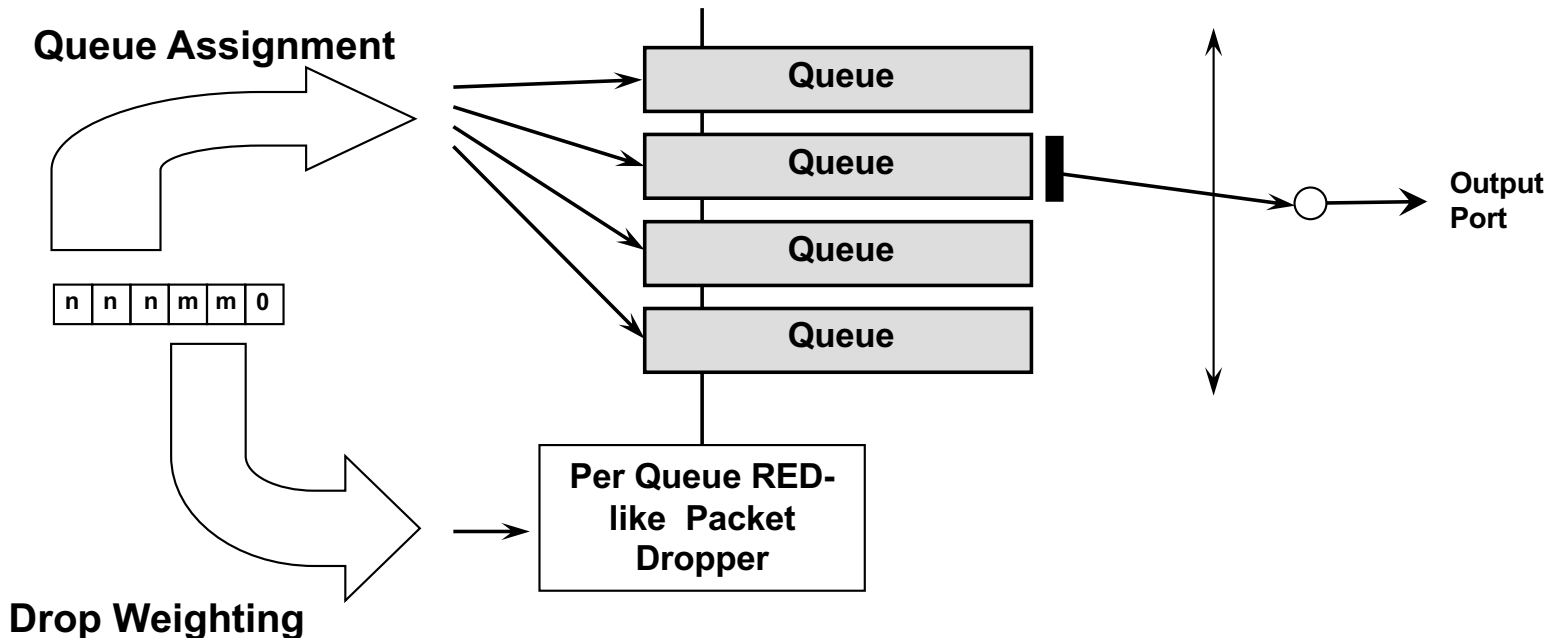
- Rate shape or police EF traffic on entry to the DS domain, to **limit the rates at which EF traffic may enter the network core**
- Configure the EF packet-servicing interval at every core router to exceed the expected aggregate arrival rate of EF traffic
- EF packet-servicing intervals must be **unaffected by the amount of non-EF traffic** waiting to be scheduled at any given instant



# DiffServ – Assured Forwarding (AF) PHB

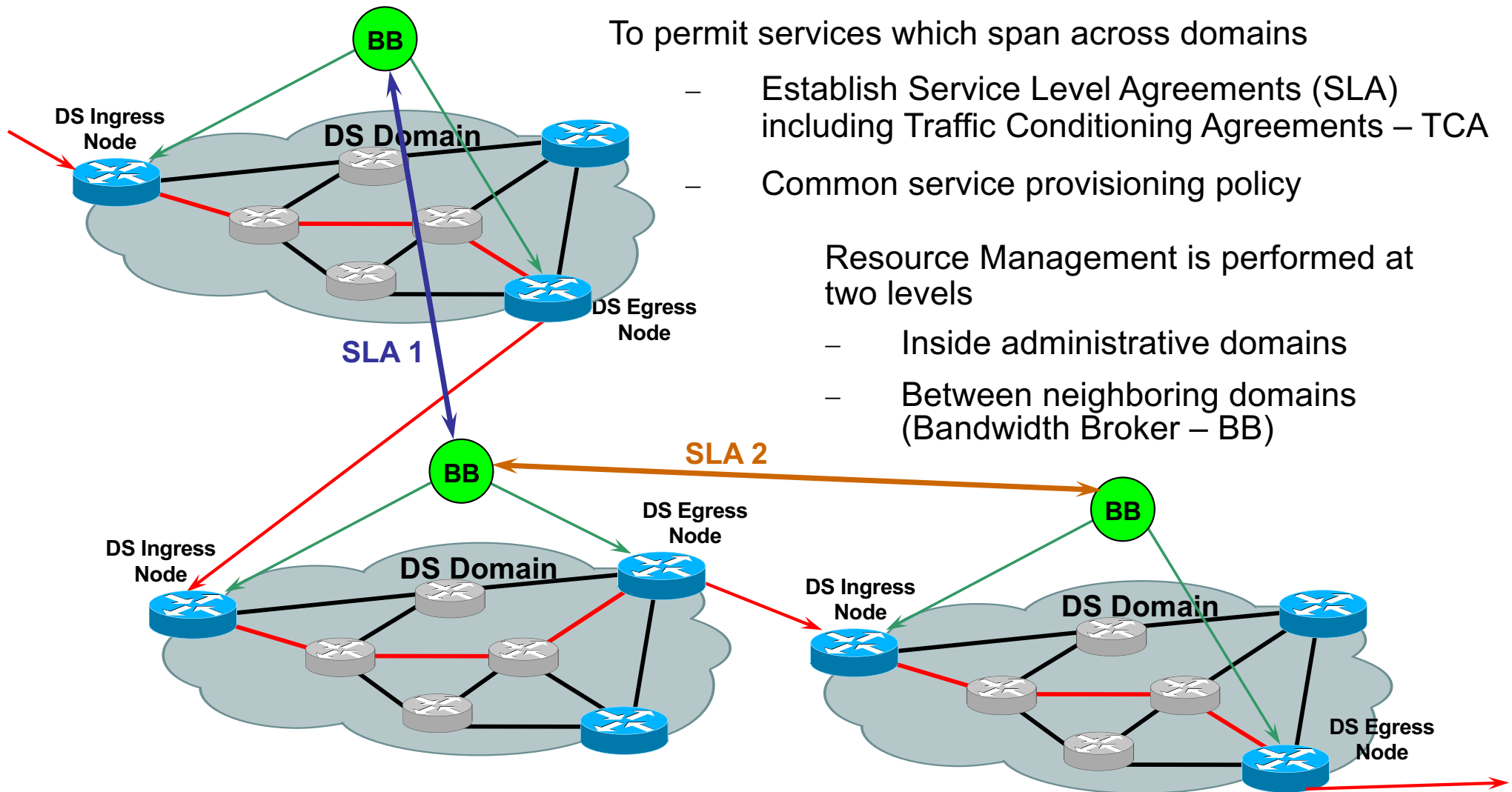
Group of PHBs for building edge-to-edge services

- Relative bandwidth availability
- Packet drop characteristics



Parameters (drop probabilities, queue sizes, scheduling parameters) are assigned by the network operator to support desired end-to-end services

# DiffServ – Two-tier Architecture



To permit services which span across domains

- Establish Service Level Agreements (SLA) including Traffic Conditioning Agreements – TCA
- Common service provisioning policy

Resource Management is performed at two levels

- Inside administrative domains
- Between neighboring domains (Bandwidth Broker – BB)

Concatenation of bilateral agreements leads to end-to-end QoS delivery paths  
But: Agreements are bilateral only!

## DiffServ – Summary

- Wide variety of services
- Easy introduction of new services in already existing DS enabled networks
- Decoupling of services from application in use
- Avoid per-microflow or per-customer state handling within core network nodes => scalability
- Interoperability with old network nodes
- Supports incremental deployment
- Division of forwarding path and management plane

# Overprovisioning

**Idea:** solve the QoS problem at deployment time (define the network in a way that overload will never occur); best effort transport

## Network design:

- Identify the maximal input to the network in each entry point
- Identify the maximum traffic for each flow from input to output
- Sum the flows for each network entity, i.e. links, switches, routers, etc., and dimension them accordingly

## Run-time: Best effort

- Ensure that the maximum traffic is not exceeded
  - by design or
  - by other traffic control measures

⇒ no overload

⇒ zero (queueing-induced) loss

⇒ no queueing

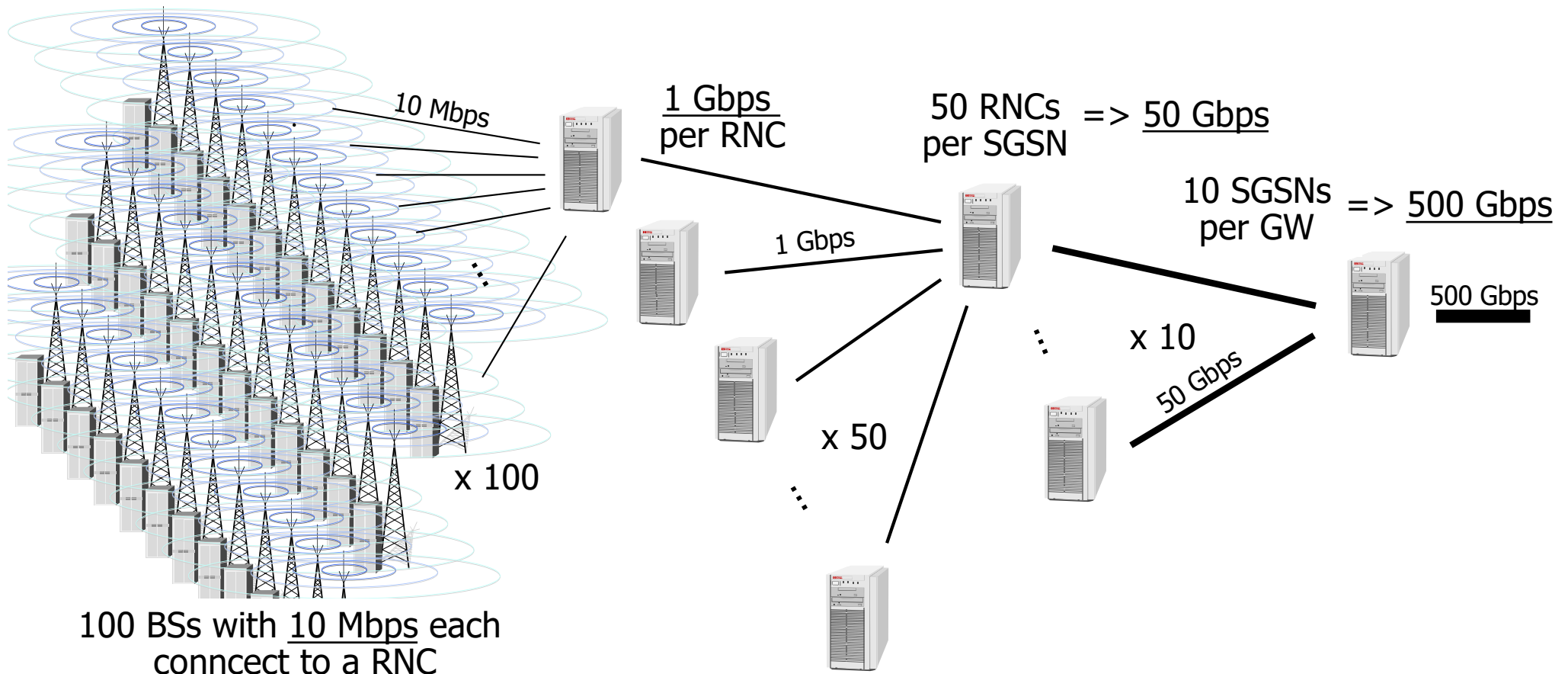
⇒ zero delay jitter

# Overprovisioning – Simple Example

## Cellular network with strict tree structure

Network design:

- Identify the maximal input to the network in each entry point
- Identify the maximum traffic for each flow from input to output
- Sum the flows for each network entity



## Wrap-up

- Best effort/  
overprovisioning
- Packet Marking
- Integrated Services
- Differentiated Services

No distinction between packets within the network (if no resources are available all packets may be queued or dropped)

Each packet is marked with a request for a type of service; nodes select routing paths and/or forwarding behaviors to satisfy the service request

First attempt of IETF to develop a service model that supports per-flow QoS guarantees; requires complex architecture along any edge-to-edge path

Minimalist counterpart to IntServ, throwing out everything that isn't essential to the provision of some aggregate service levels



# QoS Signaling

Questions to be answered:

- What's the idea behind NSIS?
- For which signaling purposes can NSIS be used?
- Does NSIS support secured signaling?
- Does it use UDP or TCP for signaling?
- What is the purpose of the GIST sublayer?



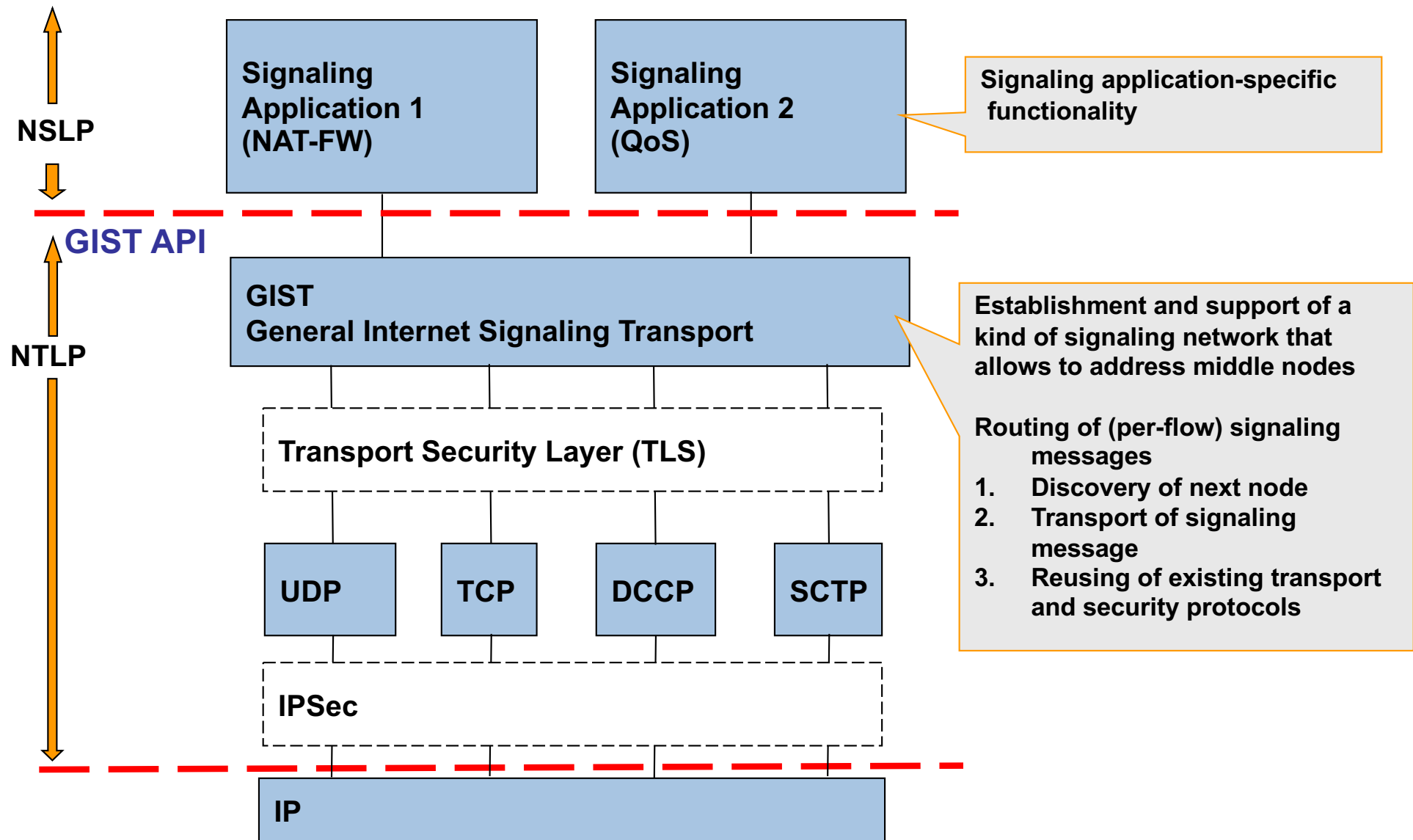
## Next Steps In Signaling (NSIS)

- Developed by the IETF nsis working group (RFC 4080)
- Framework aiming at
  - Interworking between different QoS mechanisms
  - Simplified QoS signaling
  - Support of mobility
- Same signaling problem as with RSVP is addressed
- Differences to RSVP
  - In contrast to RSVP, NSIS remains usable in different parts of the Internet without requiring a complete E2E deployment
  - Signaling can be used for purposes other than resource reservations

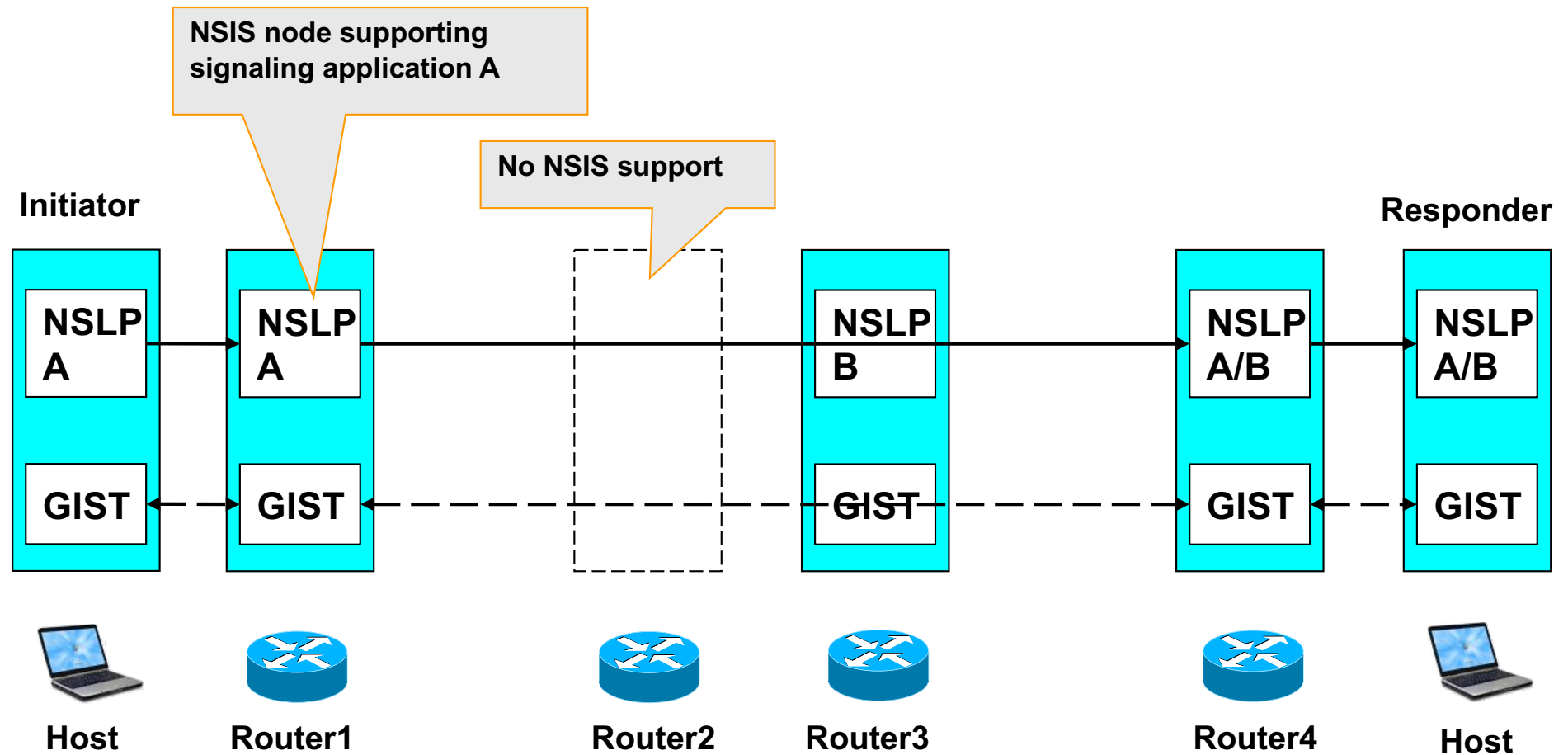
## NSIS – Overview

- NSIS aims at providing a global model that supports several signaling applications by separating the protocol stack into two layers
  - NSIS Signaling Layer Protocol (NSLP)
    - Contains different signaling applications, e.g. QoS signaling (RFC 5974), NAT/Firewall (RFC 5973), etc.
    - Communicates with NTLP
  - NSIS Transport Layer Protocol (NTLP)
    - Interface between the NSLP and IP
    - GIST (General Internet Signaling Transport protocol, (RFC 5971)
      - Common signaling transport service for different signaling applications
      - Interacts with other security and transport protocols, e.g. TCP, IPSec

# NSIS – Overview



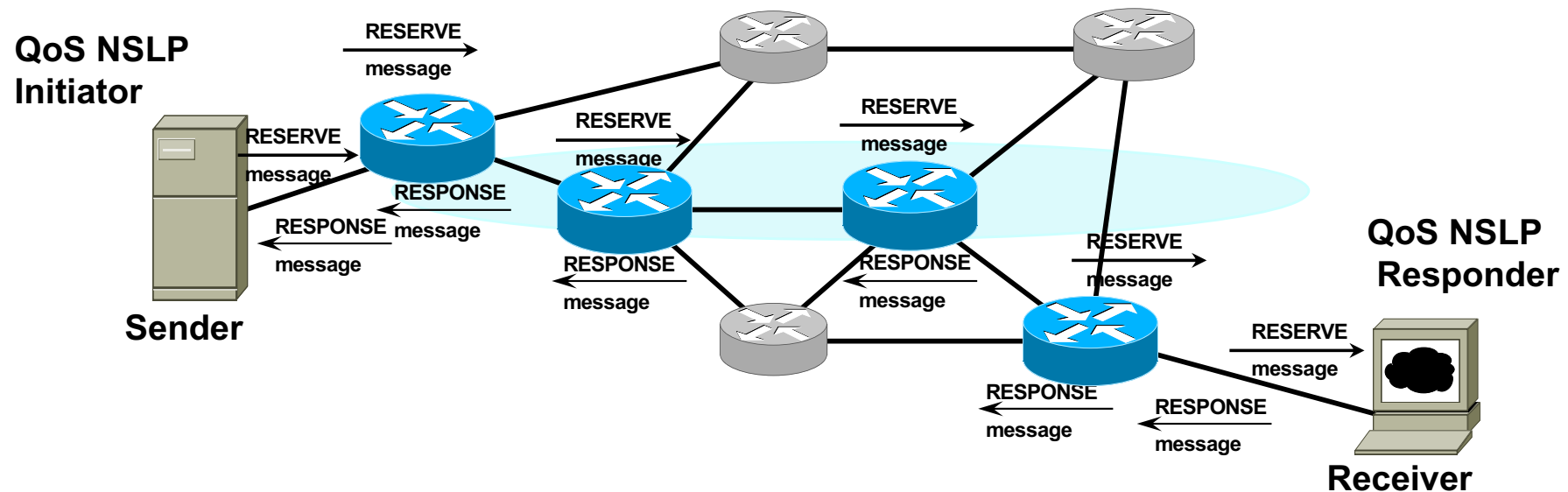
# NSIS – NTLP/NSLP Scenario



## QoS – NSLP

- RSVP-like operation, soft-state, however only unicast is supported
- Sender- and receiver-initiated reservations
- Four types of messages
  - RESERVE: creates, modifies or deletes reservation state
  - QUERY: discovers available resources along a certain path
  - RESPONSE: acknowledgement indicating reception of RESERVE or QUERY message
  - NOTIFY: notification in case of errors

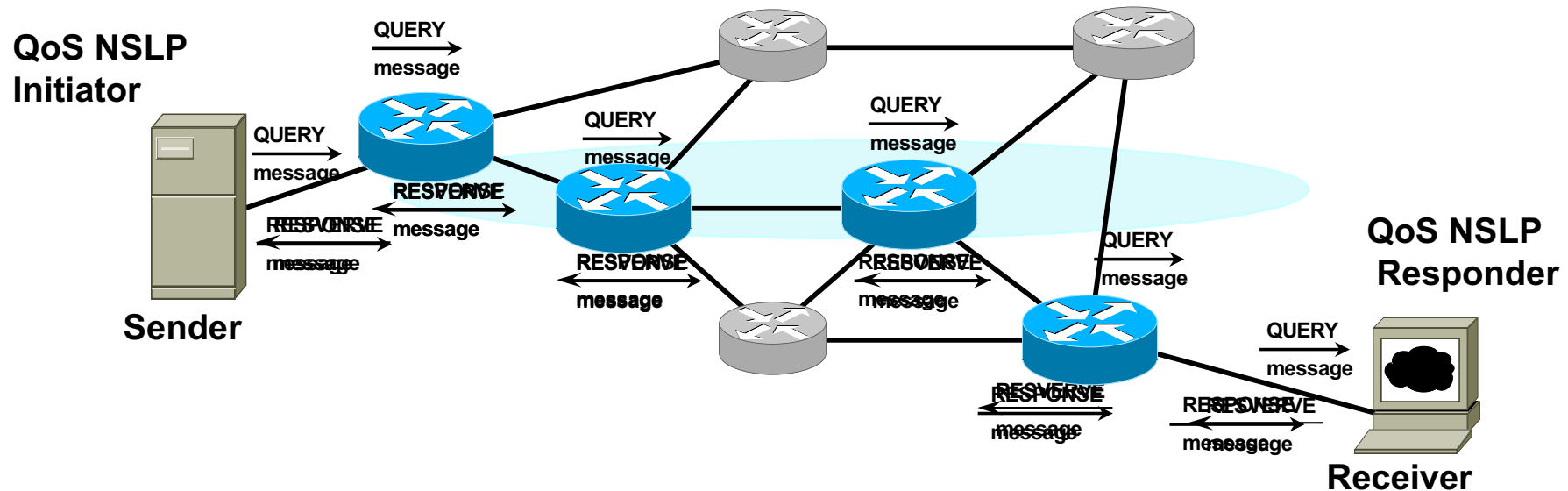
# Sender-Initiated Reservation



- (1) Sender initiates and completes the reservation issuing a RESERVE message
- (2) Receiver responds with a RESPONSE (ACK) message

Faster establishment of a reservation

# Receiver-Initiated Reservation



- (1) Sender initiates a QUERY message to inform the receiver and to prepare the network
- (2) Receiver prompts the reservation issuing a RESERVE message
- (3) Sender responds with a RESPONSE (ACK) message

Similar to RSVP mechanisms (except for the RESPONSE message)

## NSIS – Summary

- Support of different signaling applications
- Decoupling of “application” (called discovery) and transport of signaling messages
- Flexible flows, each session has an ID
  - Flow ID can be changed → support of mobility
- Receiver- and sender-oriented reservation
- Better scalability and extensibility than other mechanisms



# QoS over Wireless

Questions to be answered:

- Why is QoS considered a cross-layer issue?
- What influences the QoS of a wireless link? Can the physical layer adapt or influence the QoS? How? Are there limits of this?
- Which PHY strategies are appropriate for the maximization of the throughput?
- Which PHY strategies are appropriate for the minimization of the error rate? What are the drawbacks of the different approaches?
- What is the benefit of ...
- What would be a good strategy to optimize the system capacity of the cellular system as a whole? On which layer would this apply?

## What we have learned about network QoS so far

- Different mechanisms are employed on different parts of the end-to-end connection and different protocol layers
- Wireless is the weakest part of the connection and limits its quality & capacity
- The quality of the radio channel may vary highly over time

### How to combat this on layer 2/3?

- Application of QoS mechanisms to wireless links, i.e. radio resources
  - reservation (IntServ) => appropriate where the amount of (varying!) transport resources and the number of connections is small and the QoS requirements (esp. latency) are hard
  - differentiation (DiffServ) => appropriate where a large number of connections has to be handled or QoS requirements are moderate
  - overprovisioning => appropriate where resources are abundant (typically not true for air interface) or maximum traffic volume is known (may hold for access network)

# What we have learned from wireless PHY so far

Wireless is the weakest and most crucial part!

- **Instable QoS on wireless links due to fading, mobility, etc. caused by**
  - **Highly varying signal quality and path loss (due to none-line-of-sight, multipath, shadowing, etc.)**
  - **Highly varying interference from others**

Assuming invariant L2 & PHY settings (modulation, coding, TX power, scheduling, etc.),

the fast variation of the SINR results in high variation of the error rate,

thus, resulting in high variation of the effective throughput (see Shannon's law for impact of SNR on throughput)

$$C = B \log_2 \left( 1 + \frac{S}{N} \right) \text{ [bps]}$$

However, Shannon also tells us that we can keep the effective throughput constant if

- we keep S/N (roughly equivalent to SINR) and channel bandwidth B constant!
- or alternatively increase the used channel bandwidth B (more radio resources) if SINR deteriorates!

# How to combat varying channel quality

**What is our goals?** What are our end-to-end requirements?

- maximal throughput at minimal cost for the radio resource,
- small error rate,
- minimal latency (for small packets) or
- extremely small error rate combined with minimal latency

**How can we achieve our goals over the wireless interface?**

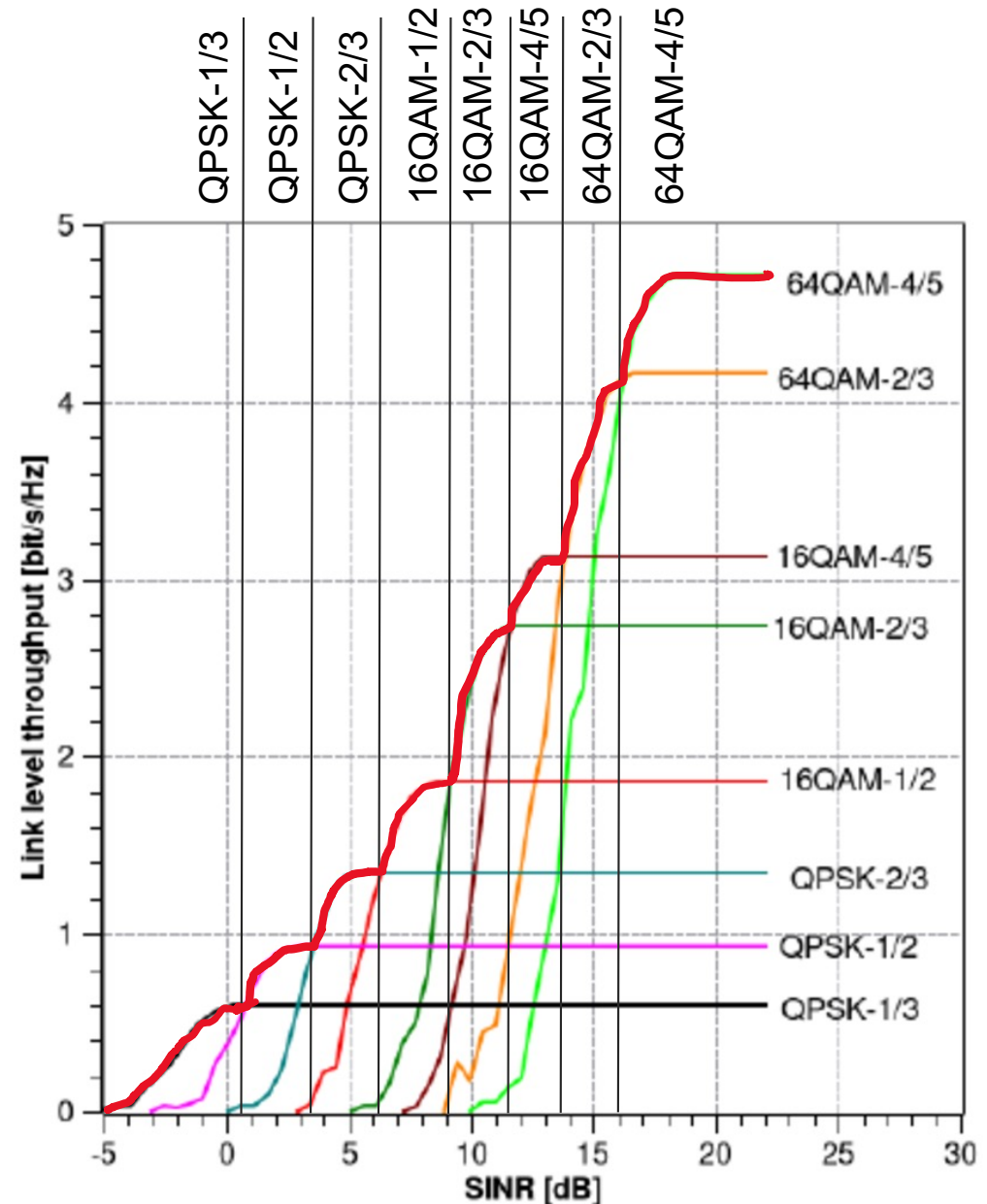
- adapt modulation (e.g. QAM => QPSK => BPSK)
- adapt coding (increase redundancy for Forward Error Correction)
- reduce packet/frame size (smaller frames have lower probability for errors)
- increase transmit power – but increases interference to others!
- change link (handover) or use multiple links simultaneously
- use multiple antennas simultaneously
- wait for better link quality (consider fast changes due to mobility)

Summarizing, depending on the specific end-to-end requirements, different measures are appropriate! => see following slides

# Wireless Throughput Optimization

## Mapping of SINR on MCS

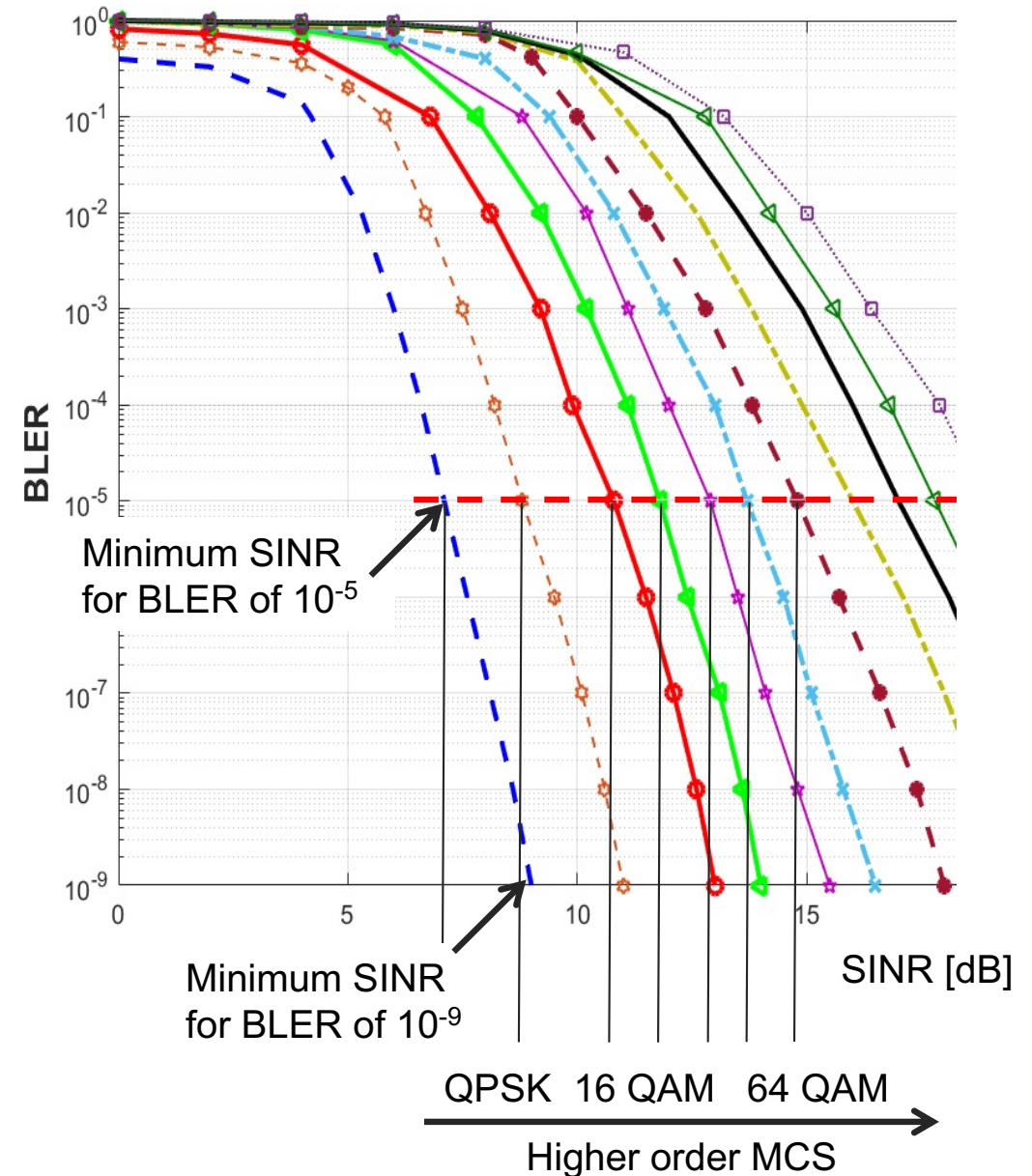
Select the MCS (Modulation and Coding Scheme) that maximizes the throughput at minimal cost to the radio system as such (red line)



# Ultra-Reliable Low Latency Communication

## Mapping of SINR on MCS for Ultra-Reliable Low Latency Communication (URLLC):

Select highest MCS that meets the given BLER target (here  $10^{-5}$ ) for the given SINR (link condition) at minimal cost



# QoS over wireless – Summary

## Wireless QoS is a moving target!

**There are lots of choices over multiple layers to choose from:**

- adapt modulation (e.g. QAM => QPSK => BPSK)
- adapt coding (increase redundancy for Forward Error Correction)
- reduce packet/frame size (smaller frames have lower probability for errors)
- increase transmit power – but increases interference to others!
- select better radio link (handover)
- use multiple radio links simultaneously
- use multiple antennas simultaneously (SU-MIMO) or beamforming
- wait for better link quality (consider fast changes due to mobility)
- apply reservations
- optimize scheduling schemes
- ...

**The best combination of selected solution depends on the specific user requirements as well as the specific radio situation!**

# QoS in 802.11

Questions to be answered:

- DCF extensions for QoS provisioning:
  - What is the idea of the DCF extension? Which QoS strategy does it implement?
  - How does the extended version of DCF work?
  - What are the parameters to differentiate traffic in EDCA?
  - Provide an example for the parameters for 4 traffic classes
  - With which parameter settings can an EDCA-enabled device outperform a non-QoS enabled device?
  - What are the limits of the DCF extension with respect to QoS?
  - What is the purpose of TXOP?
- HCCA
  - Which QoS strategy does it implement?
  - Which are the important functions for its implementation?
- How do the extensions react to changing channel conditions?



## QoS in WLANs – 802.11e

Extensions of MAC layer:

- **Hybrid Coordination Function – HCF**
  - Contention and Contention Free Periods (CP and CFPs)
- **Enhanced Distributed Channel Access – EDCA**
  - Enhanced DCF
    - Differentiation of access for different traffic classes
  - Differentiated services (→ DiffServ)
- **HCF Controlled Channel Access – HCCA**
  - Extension to PCF
    - Polling of stations in CFP
    - Provision of maximum access time to medium (TXOP)
    - Enforcement of superframes
  - Guaranteed service (→ IntServ)
- QoS-enhanced Basic Service Set (QBSS) replaces BSS

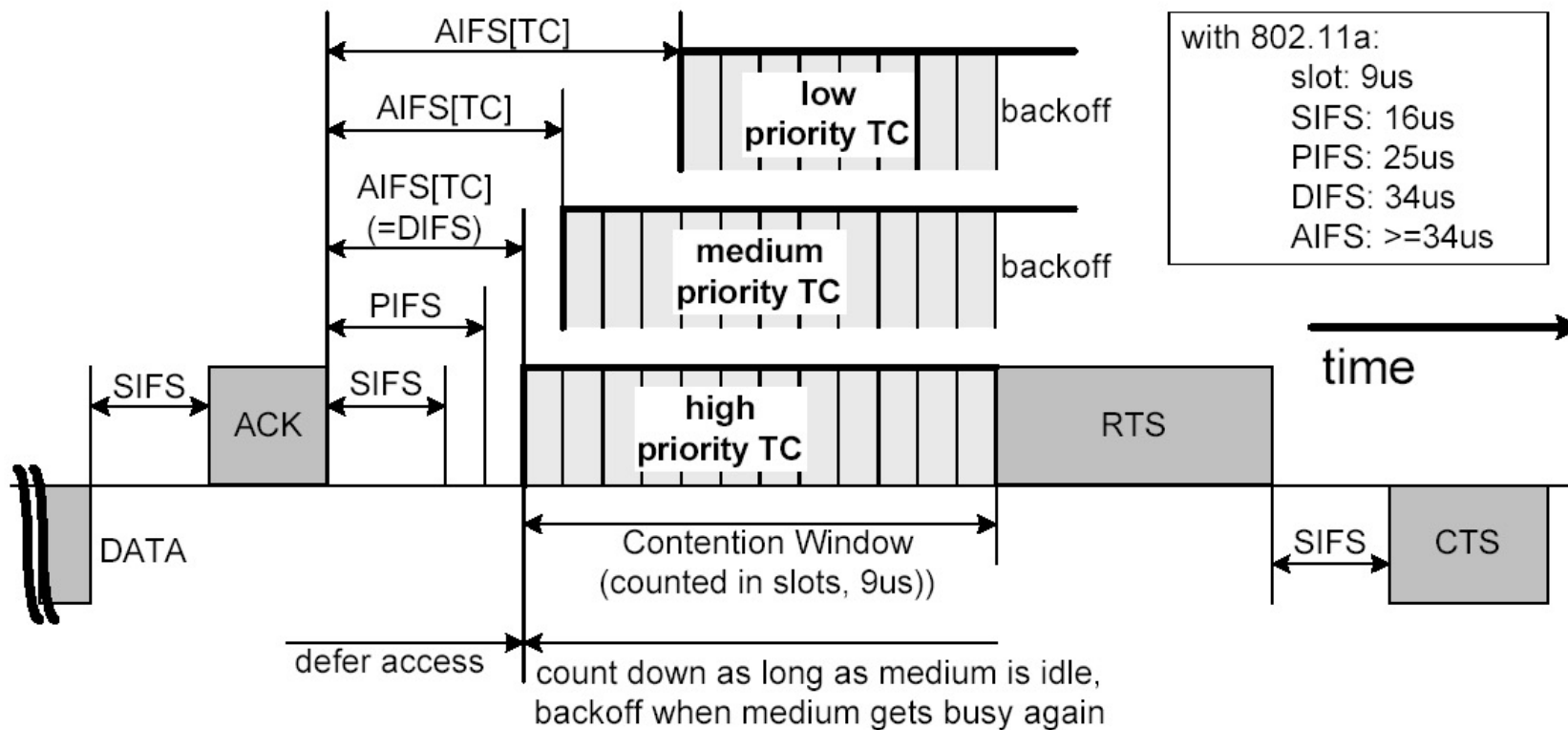
## 802.11e – EDCA (Enhanced Distributed Channel Access)

- Review of DCF (Distributed Coordination Function)
  - CSMA/CA
  - Transmits the frame directly if the medium is found idle for **DIFS** (DCF InterFrame Space)
  - Otherwise, defer the transmission and start the backoff process
  - $\text{Backoff\_time} = \text{rand}[0, \mathbf{CW}]$ ,  $CW_{\min} < CW < CW_{\max}$
  - Backoff timer decreases only when the medium is idle
  - Transmits the frame if backoff timer expires
- EDCA: Priority-based access scheme
  - Replaces DIFS with different AIFS (Arbitration InterFrame Space), depending on traffic characteristics
  - Adapts the contention window size to traffic characteristics

=> Different random backoff times and AIFSs to provide differentiated services
- The relative performance is not easy to control
  - The performance is NOT proportional to the backoff factor ratios
  - It depends on the number of contending stations

## 802.11e – EDCA

- Enhancement of access during Contention Period (CP)
- Multiple backoff instances for data streams => different priorities
- Priority over legacy stations (ensured for  $CW_{min}[TC] < 15$ )

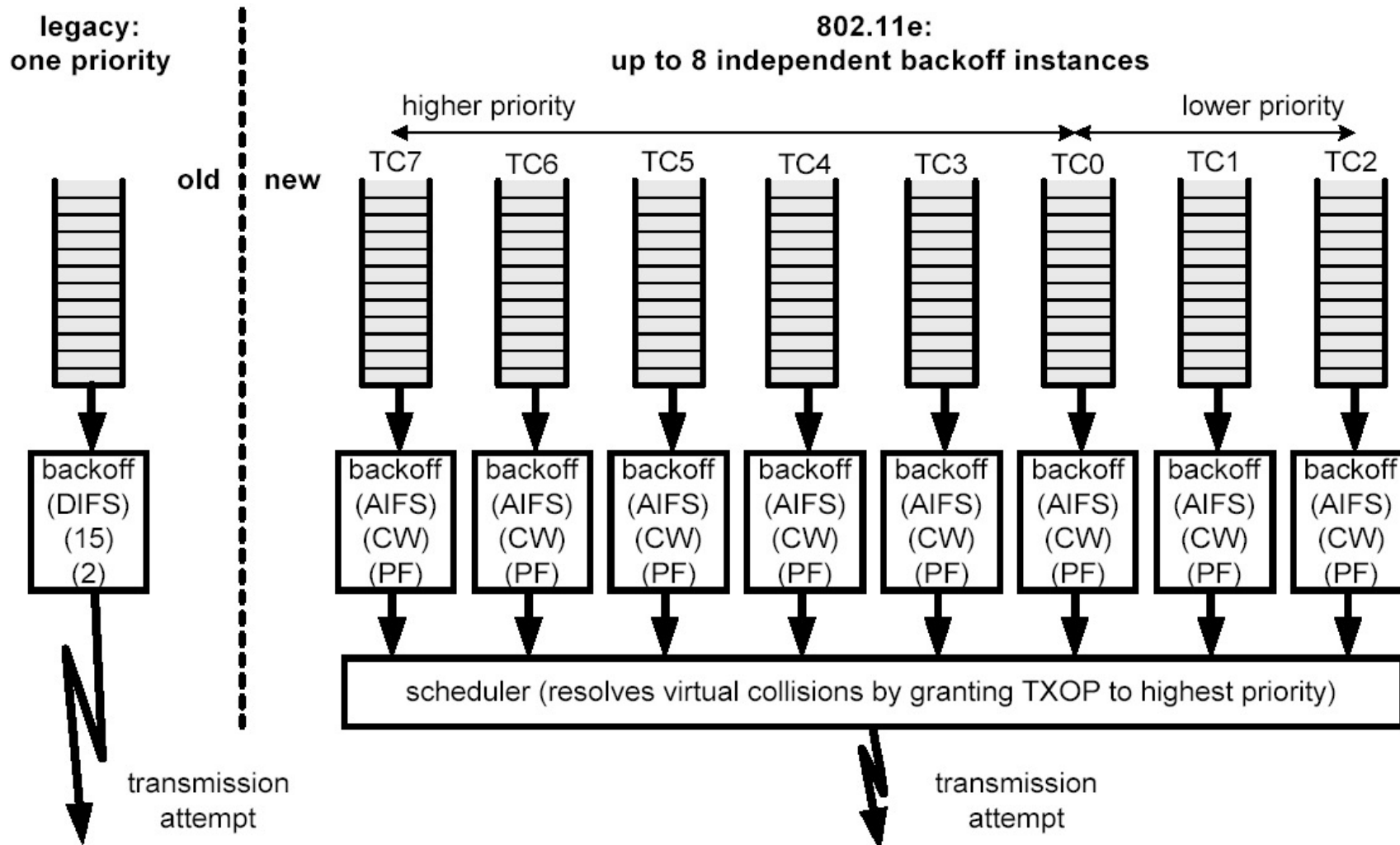


Parameters per Traffic Category (TC):

- AFIS Arbitration Inter Frame Space
- CW Contention Window (min & max values)
- TXOP Transmit opportunity (limit on access time to channel)

## 802.11e – EDCA

Up to 8 transmission queues per station



# 802.11e – WiFi-Multimedia (WMM) Specification

WiFi-Multimedia (WMM) definition supports a subset of the EDCA

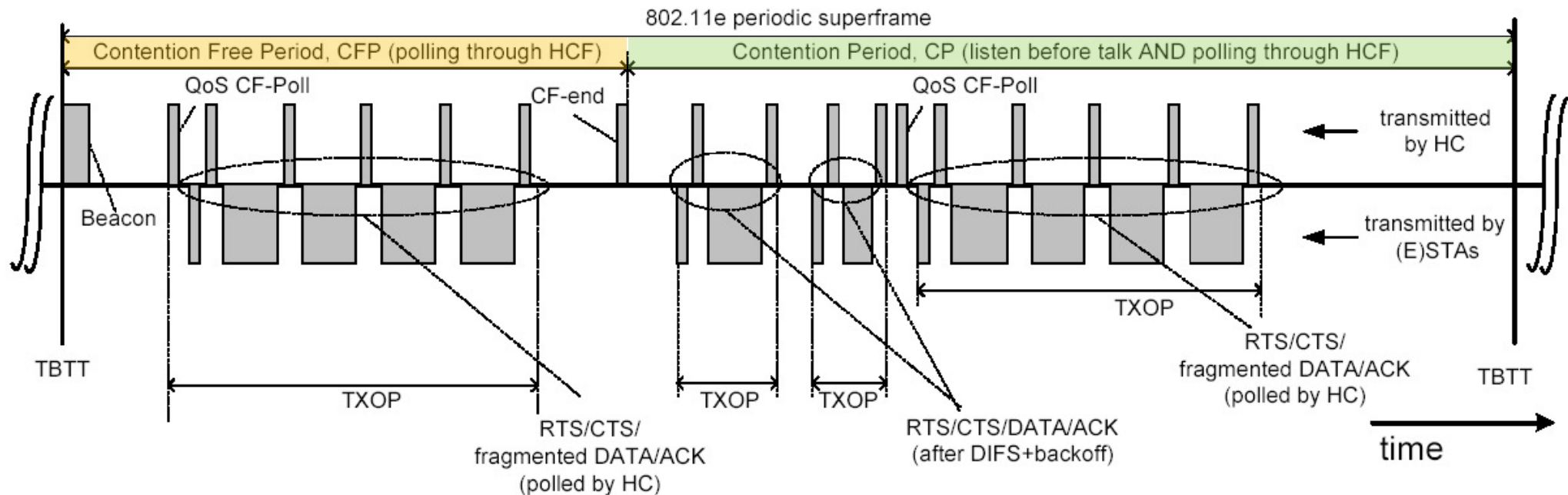
- 4 traffic classes
  - Best effort (AC\_BE)
  - Background (AC\_BK)
  - Video (AC\_VI)
  - Voice (AC\_VO)
- The traffic classes are different in
  - AIFS (IFS time)
  - Size of contention window ( $CW_{\min}$  und  $CW_{\max}$ )
  - Maximum allowed transmission time (MaxTXOP)

## 802.11e – HCCA (HCF Controlled Channel Access)

- Provides policing and deterministic channel access by controlling the channel through the HC (Hybrid Coordinator)
- Operates in CFP (Contention Free Period) and CP (Contention Period)
- Supports **IntServ**
- **Admission** (or rejection) of stations based on Traffic Specification (TSPEC)
  - min, mean & max data rate
  - delay bound
  - nominal & maximum MSDU size
  - user priority, maximum burst size
  - ...
- HC derives **schedule** to provide the guaranteed QoS requirements

## 802.11e – HCF

- Operates both EDCA and HCCA
- Includes CFP and CP phases
- Provides IntServ and DiffServ



## QoS in 802.11 – Wrap-up

- New coordination functions for channel access
  - HCF Controlled Channel Access (HCCA) replaces PCF
  - Enhanced Distributed Channel Access (EDCA) replaces DCF
- Provides services roughly comparable to IntServ and DiffServ, respectively
- Focuses on the implementation of QoS on Layer 2 (MAC)
- No adaptation of PHY layer parameters as
  - transmit power control
  - interaction with modulation and coding scheme

⇒ focus on QoS for stationary wireless systems



# QoS and Mobility

Questions to be answered:

- What is the conflict between mobility management and QoS?
- What are possible solutions to this conflict?
- What are the pros and cons of different approaches to the integration of QoS and mobility management

# How does mobility affect QoS?

QoS is dominated by the weakest part of the connection:

- High variance in quality and capacity of wireless channel due to movement of user/equipment (fading, shadowing, etc.)
- Attachment to changing infrastructure components over time due to mobility

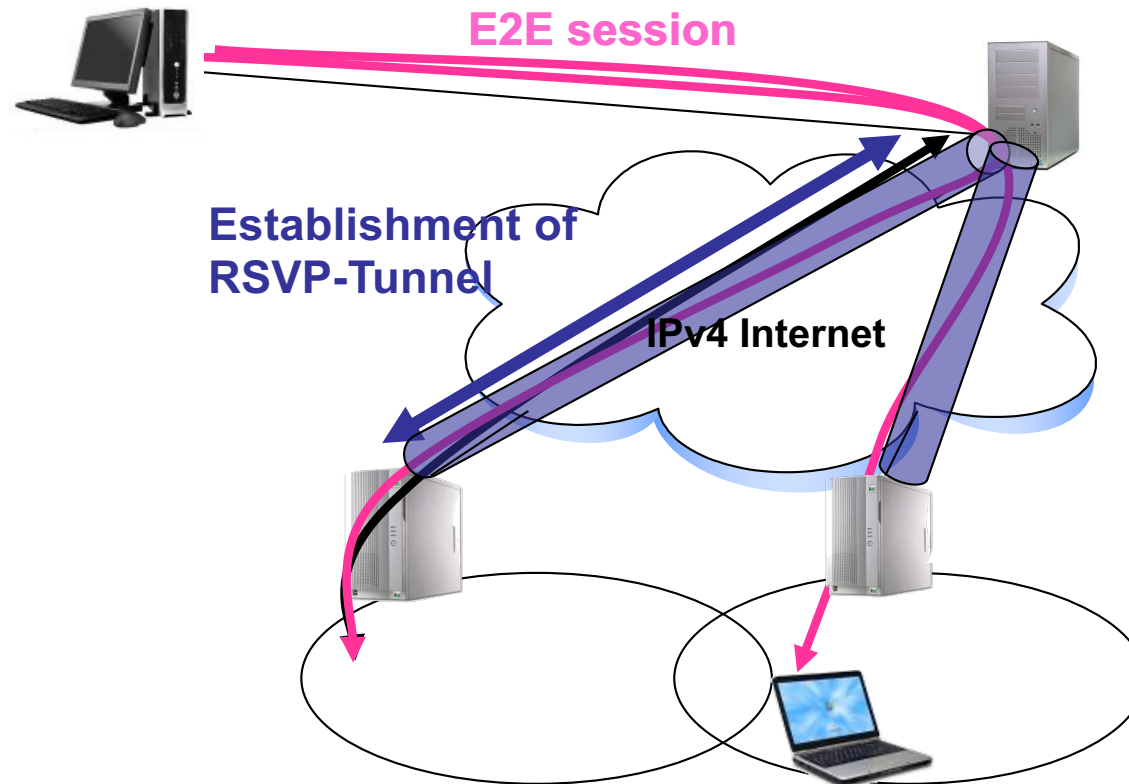
Focus here: support for mobility by QoS mechanisms to handle changes of point of access to the infrastructure

- After movements, the user has to reserve resources again
  - Availability of resources in the new location
  - Reservation latency (in addition to the handoff latency)
  - Releasing resources reserved on the old path
- Solution
  - Coupling between QoS and mobility mechanisms
  - fast reservation and release of resources

# Coupling between QoS and Mobility Solutions

- No coupling
  - Protocols work separately and independently
  - For any change in the transmission path, i.e. due to mobility, the mobile and its communication peer need to trigger QoS management
  - High delays due to end-to-end handling of QoS
- Loose coupling
  - Separate protocols for mobility and QoS, but supporting interfaces for fast interaction
  - Example: Simple QoS
- Tight integraton
  - Single joint protocol or mechanism for mobility and QoS
  - Examples:
    - Wireless Lightweight Reservation Protocol (WLRP)
    - QoS-aware Mobile IP Fast Authentication (QoMIFA)
    - QoS-handling in UMTS, LTE and 5G

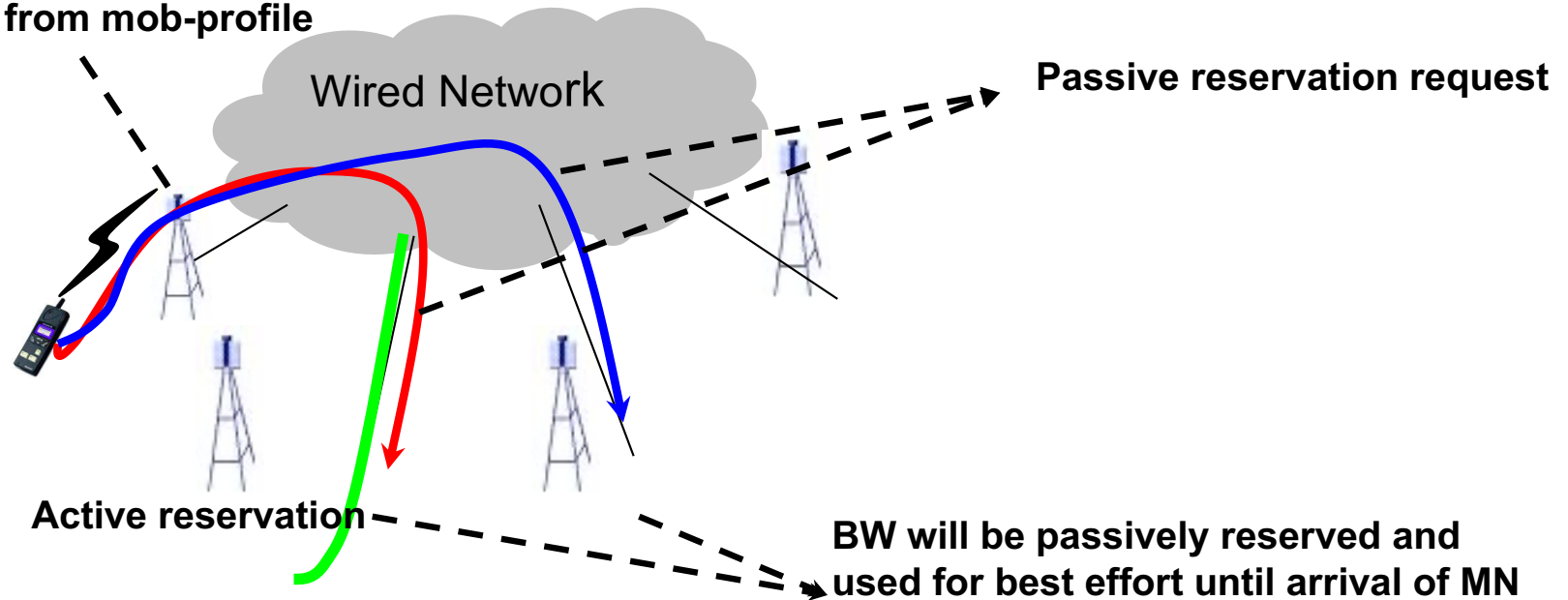
# Loose Coupling: Simple QoS



- Integrating RSVP with MobileIP
- E2E RSVP session between the CN and MN
- Additional RSVP session/tunnel between the HA and FA to offer the QoS guarantee for tunneled packets

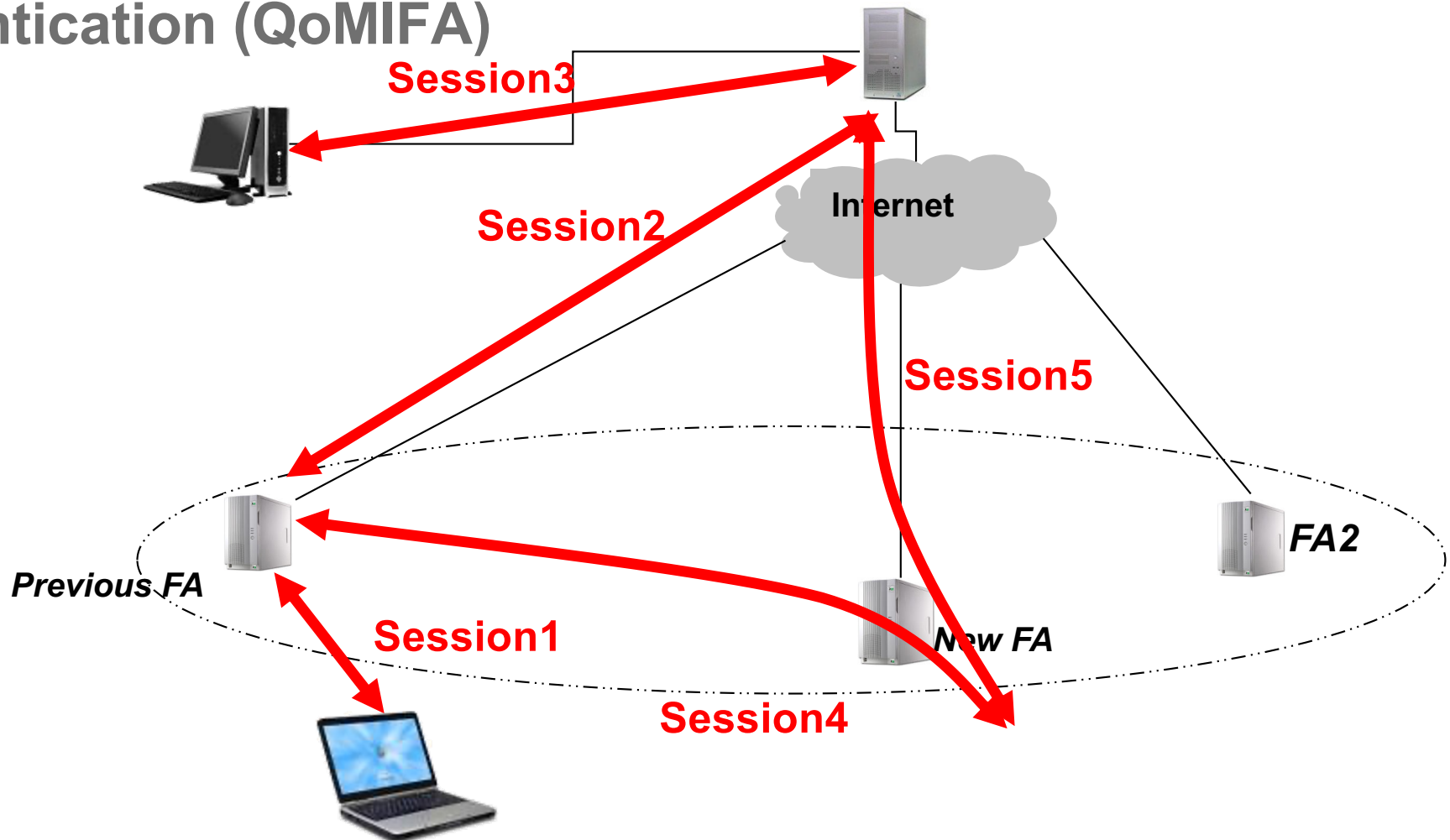
# Tight Integration: Wireless Lightweight Reservation Protocol (WLRP)

BS defines the possible cells the MN will move to from mob-profile



- MN sends reports periodically for tracking purposes
- Network defines the neighbors, where the MN may move to, from the mobility profile (mob-profile)
- Passive reservations in neighbors
- Passive reservation at a BS changes to active upon the arrival of the MN to this BS

# Tight Integration: QoS-aware Mobile IP Fast Authentication (QoMIFA)

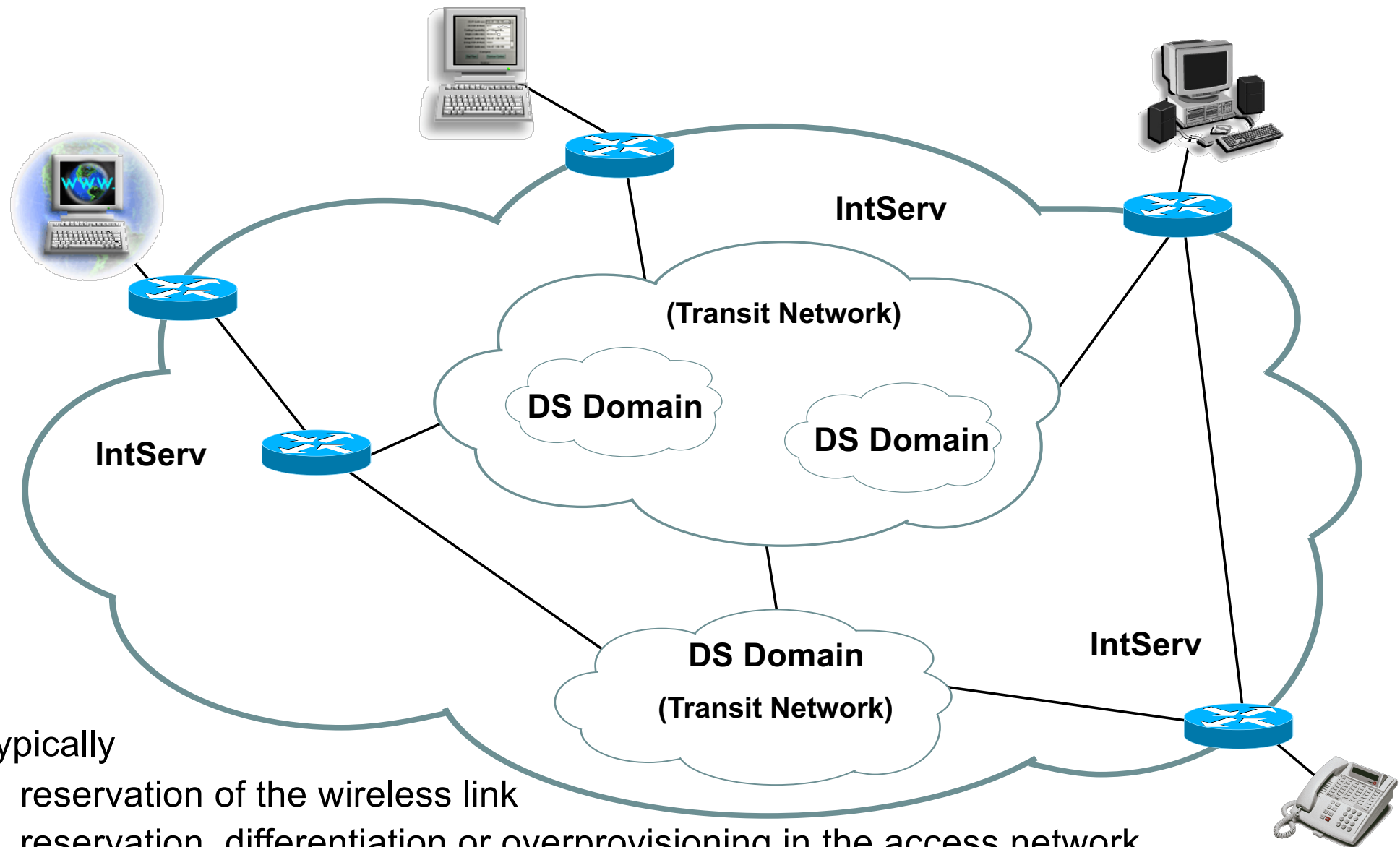


- Integrating RSVP with MIFA (Mobile IP Fast Authentication)
- Extension of RSVP through adding a new object to transport MIFA control messages
- Handoff and resource reservation are achieved simultaneously

## QoS and Mobility – Summary

- Mobility highly affects the performance of QoS mechanisms
- QoS mechanisms should interact with mobility solutions
- GSM, UMTS & LTE employ **make-before-break** strategy to jointly handle mobility and QoS proactively
- Compare to 802.11: **make-after-break** handover on the physical layer is the main reason for delay, i.e. scan for AP and reassociation with it

# Summary 1 – Mixing techniques to provide E2E QoS



Typically

- reservation of the wireless link
- reservation, differentiation or overprovisioning in the access network
- differentiation in the backbone



## Summary 2 – QoS over wireless

- Problems due to lack of proportionality between radio resource and transport resource (fading, etc.)
- QoS has to be provided end-to-end, but different mechanisms may be used on different parts of the end-to-end connection
- Application of the mechanisms to the air interface
  - Reservation (IntServ) => appropriate where the amount of resources and the number of connections is small and the QoS requirements are hard
  - Differentiation (DiffServ) => appropriate where a large number of connections has to be handled or QoS requirements are moderate
  - Overprovisioning => appropriate where resources are abundant (typically not true for air interface) or max. traffic volume is known (may hold for access network)
- UMTS provides a variety of the techniques in different parts of the system (IntServ and DiffServ); LTE is mainly focusing on DiffServ

# References

## Books on 802.11:

- F. J. Kauffels, "Wireless LANs: Drahtlose Netze planen und verwirklichen, der Standard IEEE 802.11 im Detail, WLAN-Design und Sicherheitsrichtlinien", 1. Aufl., mitp-Verl., Bonn 2002 .
- F. Ohrtman, "WiFi-Handbook – Building 802.11b wireless networks", McGraw-Hill, 2003.
- J. Schiller, „Mobile Communications (German and English)“, Kap 7.3, Addison-Wesley, 2002.

## Details on 802.11e:

- A. Lindgren, A. Almquist, O. Schelén, "Quality of service schemes for IEEE 802.11 wireless LANs: an evaluation", Mobile Networks and Applications, Volume 8 Issue 3, June 2003.
- D. Gu, J. Zhang, "QoS enhancement in IEEE 802.11 wireless local area networks", IEEE Communications Magazine, volume: 41 issue: 6, June 2003.
- Q. Qiang, L. Jacob, R. Radhakrishna Pillai, B. Prabhakaran, "MAC protocol enhancements for QoS guarantee and fairness over the IEEE 802.11 wireless LANs," in proceeding of the 11th Intl. Conference on Computer Communications and Networks, USA, October 2002.
- S. Mangold, S. Choi, P. May, O. Klein, G. Hiertz, L. Stibor, "IEEE 802.11e wireless LAN for quality of service", in proceeding of European Wireless (EW2002), Italy, February 2002.

## Web Links for 802.11:

- The IEEE 802.11 Wireless LAN Standards  
<http://standards.ieee.org/getieee802/802.11.html>
- Introduction to the IEEE 802.11 Wireless LAN Standard  
<http://www.wlana.org/learn/80211.htm>

# References

## Basics, IntServ and Diffserv

- G. Armitage, "Quality of service in IP networks: foundations for a multi-service internet", printed by Indianapolis, Ind. MTP, 2000, ISBN:1-578-70189-9.
- R. Braden, D. D. Clark, and S. Shenker, "Integrated Services in the Internet architecture: An overview", RFC 1633, June 1994.
- R. Braden, L. Zang, S. Berson, S. Herzog, S. Jamin, "Resource reservation protocol RSVP", RFC 2205, September 1997.
- K. Nichols, S. Blake, F. Baker, D. Black, "Definition of the Differentiated Services Field (DS Field) in the IPv4 and IPv6 Headers", RFC 2474, December 1998.

## NSIS

- R. Hancock, G. Karagiannis, J. Loughney, S. Van den Bosch, "Next Steps in Signaling (NSIS): Framework", RFC 4080, June 2005.
- J. Manner, G. Karagiannis, A. McDonald, "NSLP for Quality-of-Service signaling", Internet draft, February 2008.
- H. Schulzrinne, R. Hancock, "GIST: General Internet Signaling Transport", Internet draft, March 2009.

## QoS and Mobility Management

- S. Parameswaran, "WLRP: A Resource Reservation Protocol for Quality of Service in Next-Generation Wireless Networks", in proceeding of the IEEE Local Computer Networks (LCN'03), Germany, October 2003.
- A. Terzis, Mani Srivastava and Lixia Zhang, "A simple QoS signaling protocol for mobile hosts in the integrated services Internet", in proceedings of IEEE INFOCOM'99, New York, March 1999.
- E. Alnasouri, A. Mitschele-Thiel, R. Böringer, A. Diab, "QoMIFA: A QoS enabled Mobility Management Framework in ALL-IP Network", 17th Annual IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC'06), Finland, September 2006.

## UMTS

- S. Baudet, C. Besset-Bathias, P. Frêne, N. Giroux: "QoS implementation in UMTS networks", Alcatel Telecommunications Review, 1st Quarter 2001.